



Simmering Sustainability at FoodLab Alpine EPFL Restaurant

GROUP 2
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Executive Summary

This study presents a comparative **Life Cycle Assessment (LCA)** of **twelve lunch menus served at FoodLab Alpine** on the EPFL campus: six vegetarian versus six non-vegetarian meals. The objective is to compare their environmental impacts and identify practical strategies to reduce the footprint of campus catering. The study focuses on three impact categories: climate change, biodiversity and ecosystems, and human health.

The meals are treated as complete ready-to-eat lunches for one person. The functional unit is therefore defined as **one satiating midday meal meeting minimal nutritional requirements**. The assessment adopts a cradle-to-grave perspective, covering ingredient production, packaging, transport, kitchen operations, consumption, and end-of-life treatment.

The methodology follows the European Commission's **Environmental Footprint 3.1 framework**. It combines interviews with FoodLab Alpine and EPFL restauration managers, internal EPFL operational datasets, and secondary data from Agribalyse, Ecoinvent, and Cical. Because exact recipes were confidential, several modeling assumptions were required and are documented in the report.

The baseline results show that vegetarian meals generally have the lowest impacts, while beef-based meals are by far the most impactful. Beef Meatballs are the worst-performing menu, and Alaska Pollock also performs poorly, especially for biodiversity and human health. Across all meals, **ingredient production is the dominant life cycle stage**, accounting for around 80% of the total footprint, while transport and kitchen operations each contribute about 10%.

The scenario analysis then tested **five reduction levers**. Scenario 1 examined plant-based substitution of animal proteins. Scenario 2 compared meals using the Daily Food Unit (DFU), which normalizes climate impact by nutritional value. Scenario 3 assessed different cutlery systems. Scenario 4 evaluated packaging options, including incineration, recycling, and recycled materials. Scenario 5 modeled weekly menu shifts, such as adding more vegetarian days.

The scenario results show that the greatest savings come from replacing high-impact animal proteins. The **strongest reductions are observed for beef and Alaska Pollock**, while replacing chicken or pork yields smaller benefits. The DFU-based analysis adds an important nuance: low absolute CO₂eq emissions do not automatically mean strong climate performance once nutritional value is considered, since some vegetarian meals perform less favorably after normalization because of their lower nutritional density. By contrast, operational changes such as packaging and cutlery improvements have much smaller effects. At the weekly level, increasing vegetarian days can significantly reduce impacts, although the effect depends strongly on which protein is replaced.

Overall, the study shows that the strongest lever for FoodLab Alpine is **targeted menu redesign**. Priority should be given to reducing beef and imported fish while keeping meals attractive, filling, and affordable enough to be chosen in practice.

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1 Introduction and Objectives of the Study

Food systems are a major contributor to environmental pressure worldwide. According to the United Nations, around one third of human-caused greenhouse gas emissions are linked to food systems [15]. Within this sector, animal-based foods are generally associated with higher impacts, especially red meat, dairy products, and some seafood products. This makes food choice an important lever for reducing climate impacts.

This issue is also relevant at EPFL, where food is identified as the third-largest source of greenhouse gas emissions on campus, mainly due to meat consumption. EPFL’s food-service strategy already includes one vegetarian day per week, at least 50% vegetarian daily menu items, and the goal of reaching 80% vegetarian or vegan dishes by 2030 [4]. In addition, the MegaBites project shows that EPFL is already using food-system data to identify ways of reducing the carbon footprint of campus meals [5]. This provides the institutional context for the present study, which focuses specifically on the environmental performance of selected FoodLab Alpine menus using Life Cycle Assessment.

Building on this context, this study examines FoodLab Alpine, one of EPFL’s campus restaurants, as a case study. For the final project of the course Environmental Assessment and Reporting, we conducted a comparative Life Cycle Assessment of 12 Alpine-style lunch menus, consisting of six vegetarian and six non-vegetarian. The study was carried out by Mia Chambat, Micha Streuli, Rania Ben Hamidane, and Janina Spielkamp, all students of the Master in Sustainable Management and Technology at EPFL. The assessment covers the full meal system, from ingredient production and procurement to preparation, service, and waste treatment.

The objective of the study is not only to confirm that meat-based meals tend to have higher impacts, but also to identify which reduction levers matter most in the EPFL context. These include changing protein sources, replacing meat and fish components with plant-based alternatives, improving procurement choices, optimizing kitchen operations, and reducing food waste. Since menu choice is also influenced by consumer behaviour, the study considers how vegetarian options and better menu design could support lower-impact food choices.

Overall, this LCA evaluates whether shifting FoodLab Alpine meals toward more vegetarian, lower-impact, and better-designed options can reduce food-related environmental impacts while accounting for nutrition, procurement, kitchen operations, service, and waste.

2 Literature Review

Previous research shows that Life Cycle Assessment (LCA) is a suitable method for comparing the environmental impacts of meals and diets. Heller et al. argue that diet-level LCAs should not only compare foods by mass, but also consider nutritional quality, since the choice of functional unit can strongly influence the results [9]. This is relevant for the present study, where meals are compared both by their absolute environmental impact and by their climate impact per Daily Functional Unit (DFU) [3].

Research on institutional meals also shows that menu composition can strongly influence environmental performance. Martinez et al. assessed school lunch menus following Spanish dietary guidelines and found that school meals can be evaluated through carbon footprint calculations while still considering nutritional requirements [13]. This supports the approach used in this study, where FoodLab Alpine menus are compared not only as individual dishes, but also as nutritionally relevant meals served in an institutional setting.

Overall, the literature suggests that reducing animal-based ingredients and improving menu design can be important strategies for lowering food-related impacts. However, such changes should still consider nutritional value, consumer acceptance, and practical implementation in

the kitchen. For this reason, this study combines LCA results with DFU-based analysis and scenario comparisons to evaluate which reduction strategies are most relevant for FoodLab Alpine.

3 LCA Methodology

3.1 Discussion with stakeholders

To anchor our exploratory analysis, we conducted interviews with two key figures in the campus dining ecosystem: Jean-Marc Sartori, Restaurant Manager at Alpine Food-Lab, and François Lauwick, the EPFL Restauration Manager who oversees all three campus caterers. Mr. Sartori gave us an initial look into their supplier and KPI databases, though we will need to approximate meal compositions moving forward since their exact recipes are confidential. A central theme from this conversation was that Alpine Food-Lab prioritizes meal diversity and flavor rather than low prices. However, they face a steep uphill battle with student habits: sales drop significantly on vegetarian days, largely because the student body consistently favors simple, filling comfort food like chicken nuggets, regardless of the associated price tag. Sartori noted that good plant-based cooking is a delicate science, stating, "Vegetarian food is like baking; you need to measure ingredients and assemble them artfully."

Moving up to the institutional level, our meeting with François Lauwick provided a broader look at the strict regulations and tracking databases that govern EPFL's entire dining network. Mr. Lauwick echoed the financial and culinary hurdles of moving away from meat, noting bluntly that "true, high-quality vegetarian cuisine is expensive". This financial reality is a major roadblock to expanding sustainable options without driving up costs for students.

Finally, Mr. Lauwick shed light on the broader social responsibilities of campus dining, specifically addressing why EPFL avoids standard student meal plans, which would be an interesting lever to increase the consumption of vegetarian meals (eg. cheaper meal plans if more vegetarian meals are consumed). From an administrative standpoint, a universal meal plan creates inherent inequalities, as rigid prepaid structures can inadvertently burden lower-income students. Ultimately, these interviews highlight that any future sustainability initiatives on campus cannot just focus on carbon footprints; they must actively navigate student preferences for hearty meals, the high cost of quality vegetarian ingredients, and the necessity of campus equity.

These exchanges provided both quantitative inputs (more detail in 3.4), such as supplier and KPI database information, and qualitative insights into the practical challenges of promoting vegetarian meals on campus.

3.2 Function, functional unit and reference flows

The core of this study is to understand the environmental footprint of what we eat on campus. By analyzing 12 specific Alpine lunch menus (6 vegetarian and 6 non-vegetarian) provided by the SV group, we are measuring the total impact of one person's meal from the farm to the fork and the bin (cradle-to-grave).

Function. The primary function of the system is to provide a nutritionally balanced and satiating midday meal to an individual on the EPFL campus. By defining the function through the lens of satiety, we acknowledge that a meal's value isn't just its weight, but its ability to provide the energy and satisfaction required to sustain a full day of academic or professional activity.

Functional unit. To ensure a scientifically sound "apples-to-apples" comparison between vegetarian and non-vegetarian menus, the functional unit is defined as: "The provision of one

satiating, ready-to-eat midday meal to one person on the EPFL campus, meeting minimal nutritional requirements."

Reference flows. To fulfill this functional unit, we have identified several flows of materials and energy.

Table 1: System Reference Flows per Functional Unit

Life Cycle Stage	Reference flow	Unit
Ingredients	Ingredient mass, scaled per recipe	g/meal
Packaging	Packaging mass by material type	g/meal
	Reusable transport box allocation	g/meal
Food transport	Refrigerated supplier-to-EPFL transport activity	t·km/meal
	Non-refrigerated supplier-to-EPFL transport activity	t·km/meal
	Air freight transport activity	t·km/meal
	Airport-to-EPFL road transport activity	t·km/meal
Employee commuting	Public transport activity	pkm/meal
	Passenger car distance	km/meal
	Electric bicycle distance	km/meal
	Motorcycle/scooter activity	pkm/meal
Kitchen operations	Food preparation water use	kg/meal
	Cooking electricity use	kWh/meal
	Refrigeration electricity use	kWh/meal
	Water electricity use	kWh/meal
	Dishwashing water use	kg/meal
	Dishwashing electricity use	kWh/meal
	Dishwashing detergent	g/meal
End-of-life	Preparation waste treatment	g/meal
	Plate waste treatment	g/meal
	Wastewater treatment	m ³ /meal
	Packaging waste treatment	g/meal
	Tableware/cutlery disposal	g/meal

The key parameters influencing the results are the ingredient composition of each meal, particularly the type and quantity of animal-based proteins, transport distances, employee commuting assumptions, cooking and refrigeration electricity consumption, water use, and food waste generation rates.

3.3 System and system boundaries

Figure 1 below presents the system boundaries considered in this study. A cradle-to-grave approach was adopted, covering six major life cycle stages associated with meal provision at Alpine Restaurant: ingredient production, processing and manufacturing, packaging, transportation, consumption and kitchen operations, and end-of-life. This boundary captures the main sources of environmental impact and enables a consistent comparison between menu options.

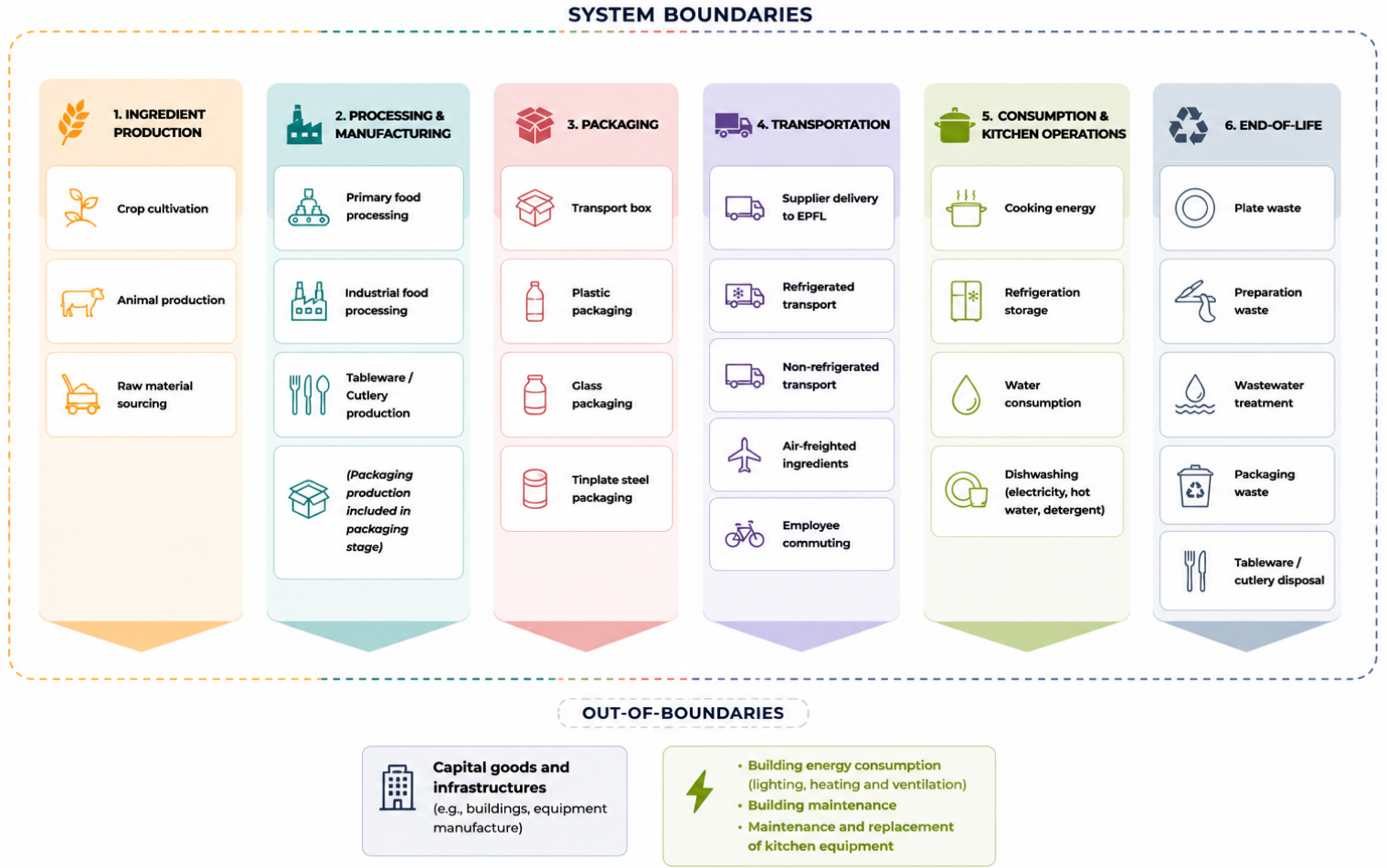


Figure 1: System boundary diagram from cradle-to-grave.

The system boundaries focus on processes directly related to meal provision and consumption at Alpine Restaurant. Capital goods associated with upstream production stages, such as agricultural machinery, industrial facilities, and manufacturing equipment, were excluded. Infrastructure-related processes at the restaurant level, including building construction, building maintenance, kitchen equipment manufacturing and replacement, as well as general building energy consumption for lighting, heating, and ventilation, were also excluded. These processes are not specific to individual meals and reliable allocation data were not available. Their contribution is therefore assumed to be similar across all menu options and unlikely to affect the comparative conclusions of the study.

3.4 Data sources

We use the Agribalyse and Ecoinvent datasets provided for the project and included on the modeling platform SimpleLCA. Most flows already include the resource acquisition, manufacturing and transport to France or retailers for Agribalyse or Ecoinvent respectively. You can access the data via <https://doc.agribalyse.fr/documentation-en/agribalyse-data/data-access> or check the main registry at <https://ecoinvent.org/database/>. For the DFU analysis, nutritional values were obtained from the Ciquial Food Composition Database [2].

In addition, all impacts throughout this study are expressed using the European Commission's Environmental Footprint (EF 3.1) methodology. This framework allows us to evaluate a comprehensive suite of indicators, including climate change, biodiversity impacts, and human

health metrics, using standardized, scientifically validated characterization factors.

Finally, following the interviews, François Lauwick provided several internal datasets that supported the modelling work. These included the FoodLab Alpine takeaway dataset, which contains daily records of takeaway meal sales at the restaurant [10]; the vegetarian and non-vegetarian meal sales dataset, which records daily meal counts by menu category [12]; and the SV Group ingredient sourcing and transport dataset, which provides information on product origins, transport modes, and supplier logistics [11]. Together, these datasets provided valuable operational information that complemented the qualitative insights obtained during the interviews.

3.5 Distribution

3.5.1 Packaging

Many of the selected Agribalyse food-product datasets used for this LCA already include typical packaging systems, including treatment at the end-of-life. However, raw ingredients do not. The impact of packaging makes up less than 1% of the total impact (EF 3.1 single score). Since adding separate packaging flows for each ingredient would risk double-counting and the impact of packaging is minor per meal, we created a separate LCA for different types of packaging.

3.5.2 Transportation

The Agribalyse food-product datasets include transportation and distribution processes representative of products consumed in France. According to the Agribalyse methodology, this generally covers the transport of raw materials to processing facilities as well as the distribution of finished products to retail outlets. Likewise, Ecoinvent production datasets include the upstream transportation activities required within the modeled supply chain. Consequently, only transportation stages specific to the Alpine restaurant system and not already represented in the selected datasets were added separately. Three additional transport components were therefore modelled: final supplier delivery to EPFL, air freight for specific ingredients, and employee commuting.

a) Food Supply Transport

Regular supplier deliveries were modelled to capture the final transport of ingredients from suppliers to EPFL. To estimate the delivery distance used in the model, a weighted average distance was calculated using the EPFL catering delivery data from 2022 [6]. The calculation was based on the distance of each supplier to EPFL, weighted by its estimated contribution to the total delivered mass. The detailed calculation is provided in Appendix A.1:

$$d_{\text{avg}} = \frac{\sum_i (\text{mass share}_i \times d_i)}{\sum_i \text{mass share}_i} = 25 \text{ km}$$

where i represents each supplier and d_i is the distance between supplier i and EPFL. This resulted in an average final delivery distance of approximately 25 km, which is applied to model the transport from suppliers or retailers to EPFL.

For each recipe, ingredients are separated into refrigerated and non-refrigerated transport flows. Refrigerated transport is used for ingredients requiring cold-chain delivery and is represented by road freight transport by lorry with a refrigeration unit. Non-refrigerated transport is used for the remaining ingredients and is represented by standard non-refrigerated road freight transport by lorry.

The transport activity is calculated separately for refrigerated and non-refrigerated ingredients as follows:

$$\text{Transport activity}_j = \frac{\sum_k m_k}{1,000,000} \times 25$$

where j represents the transport type, either refrigerated or non-refrigerated, and k represents the ingredients assigned to that transport type. The mass m_k is expressed in grams, so the division by 1,000,000 converts grams into tonnes. The result is expressed in tonne-kilometres.

b) Air-Freighted Ingredients

A separate air-freight model was added for ingredients specifically reported by Alpine Food-Lab as being transported by plane, based on the SV Group dataset provided by François Lauwick, EPFL Restauration Manager [11]. These ingredients include mint, coriander, rosemary, and oregano. Mint, coriander, and rosemary are assumed to originate from Morocco, while oregano is assumed to originate from Turkey.

For these ingredients, transport is modelled in two stages. First, air freight was modelled from the country of origin to Geneva Airport. Second, road transport is modelled from Geneva Airport to EPFL. When the ingredient required cold-chain delivery, both the air transport and the airport-to-EPFL road transport are modelled using refrigerated transport processes. For ingredients not requiring refrigeration, standard non-refrigerated transport processes are used.

The air transport activity is calculated as:

$$\text{Air transport activity}_k = \frac{m_k}{1,000,000} \times d_{\text{origin, Geneva}}$$

The airport-to-EPFL transport activity is calculated as:

$$\text{Airport-to-EPFL transport activity}_k = \frac{m_k}{1,000,000} \times d_{\text{Geneva, EPFL}}$$

where k represents the ingredient transported by air, m_k is the ingredient mass in grams, $d_{\text{origin, Geneva}}$ is the air distance between the origin airport and Geneva Airport, and $d_{\text{Geneva, EPFL}}$ is the road distance between Geneva Airport and EPFL. The road distance from Geneva Airport to EPFL is estimated at 56 km.

c) Employees Mobility

Finally, Employee commuting is included to account for the transportation impacts associated with the staff required to prepare and serve meals at the Alpine restaurant.

The average number of meals served per day was calculated using the SV Group takeaway dataset for Alpine during the week of 23/02/2026, provided by François Lauwick, EPFL Restauration Manager [10]. Based on this dataset, the restaurant served 407.6 meals per day on average.

Furthermore, based on an interview conducted with the FoodLab Alpine manager, we use a workforce number of 8 employees per day.

Based on the 2025 EPFL Mobility Survey, EPFL staff travel an average of 22.1 km per trip to campus, corresponding to a daily round-trip distance of 44.2 km [7]. The commuting impact was therefore allocated per meal using:

$$\text{Commuting distance per meal} = \frac{N_{\text{employees}} \times D_{\text{round-trip}}}{N_{\text{meals served}}}$$

where $N_{\text{employees}}$ is the number of employees working at Alpine, $D_{\text{round-trip}}$ is the average daily commuting distance, and $N_{\text{meals served}}$ is the average number of meals served per day. The commuting distance allocated to one meal is:

$$\frac{8 \times 44.2}{407.6} = 0.868 \text{ pkkm/meal}$$

This value was then distributed across transport modes using the staff modal shares obtained from the 2025 EPFL Mobility Survey: 41% public transport, 37% car, 17% bicycle, 4% walking, and 2% motorcycle/scooter [7]. Walking is neglected due to its negligible environmental impact.

$$\text{Mode-specific transport activity} = 0.868 \times \text{Mode share} \quad (1)$$

resulting in:

- Public transport: $0.868 \times 41\% = 0.356 \text{ pkkm/meal}$
- Car: $0.868 \times 37\% = 0.321 \text{ km/meal}$
- Bicycle: $0.868 \times 17\% = 0.148 \text{ km/meal}$
- Motorcycle/scooter: $0.868 \times 2\% = 0.017 \text{ pkkm/meal}$

In Ecoinvent, public transport is represented using a tram transport process as a proxy for the Lausanne metro system due to their comparable energy consumption characteristics. Car commuting is represented using an average passenger car process, bicycle commuting using an electric bicycle process, and motorcycle commuting using an average motor scooter process.

3.6 Consumption

3.6.1 Justification for using low voltage (LV)

For the life cycle inventory, a Low Voltage (LV) electricity dataset was intentionally selected to align the system boundary precisely with the point of actual consumption. In grid management frameworks, any supply below 1000V is classified as low voltage. This directly encompasses the standard 230V sockets and 400V three-phase connections that power professional kitchen equipment.

3.6.2 Water use and associated energy

Water consumption is modeled dynamically based on the total mass of the meal. Following the key performance indicators (KPIs) established by the SV Group, the total volume of water required is calculated by multiplying the weight of the meal by a factor of 3.2.

In addition to direct water consumption, the model accounts for the fixed electrical overhead tied to water delivery and sanitation. Regardless of the specific meal type, a baseline of 0.1745 kWh is applied across all calculations. This baseline encompasses:

- Building pumping: The energy required by booster pumps to deliver approximately 3.8 liters of water to the kitchen for preparation, parboiling, and dishwashing (0.0015 kWh).

- Industrial dishwashing: The energy required to heat 2.5 liters of sanitation water to 60°C for cleaning plates and cutlery (0.1730 kWh).

3.6.3 Refrigeration storage

To capture the true operational reality of a high-traffic commercial kitchen, the model incorporates a standardized 48-hour storage window in a walk-in cooler. This accounts for the standard practice of moving ambient ingredients (such as tomatoes, peppers, and onions) into cold storage overnight to prevent spoilage and pest issues.

The total refrigeration load is calculated using a standardized 48-hour energy factor of 0.00035 kWh per gram of food. This factor is comprehensive and integrates three main thermodynamic phases:

- The pull-down load: The initial energy surge required to drop ambient ingredients from room temperature (approximately 20°C) down to a stable storage temperature of 4°C.
- Steady state storage: The lower, continuous maintenance energy needed to counteract ambient heat leakage through the insulated walls of the cooler over 48 hours.
- Infiltration and traffic: An operational efficiency "tax" of 20% to 30%, which accounts for the frequent opening and closing of walk-in doors by staff during preparation shifts.

The total impact is calculated by multiplying the raw meal weight in grams by this standardized factor.

3.6.4 Cooking appliances and allocation

Energy consumption during active cooking is allocated per serving based on the specific equipment used:

- Stovetop cooking: Modeled around a large stockpot or rondeau with a standard batch capacity of 20 servings. The commercial burner is assumed to draw an average of 2.5 kW, which balances the initial high heat required for boiling with a subsequent low simmer. The per-serving energy is calculated as:

$$\text{Energy per serving (kWh)} = \frac{\text{Cooking time (h)} \times 2.5 \text{ kW}}{20 \text{ servings}}$$

- Oven cooking: Based on a baseline capacity of 60 servings per oven cycle, yielding an energy consumption rate of 0.0867 kWh per serving per hour (derived from a 5.2 kWh total draw). For complex meals where separate ingredients require individual cooking cycles, the capacity is adjusted accordingly (e.g., 30 servings per oven for two separate elements, or 20 servings for three elements).
- Deep-fryer cooking: Modeled using an industrial deep fryer drawing 3.0 kW with a standard batch size of 5 servings. The allocation follows the cooking time in hours multiplied by the 3.0 kW draw, divided by the batch size.

3.7 Waste

4 different types of waste are considered in this LCA: preparation waste, plate waste, packaging waste, and wastewater. The meals include preparation waste, plate waste, and wastewater whereas packaging waste is looked at separately.

3.7.1 Preparation Waste, Plate Waste, and Wastewater

Management of the canteen could not provide specific data on waste in the canteen due to confidentiality. However, research showed that SV Group reported 25 grams of food waste per meal in 2023 for ETH Zurich locations [14]. Applying that to Alpine restaurant and assuming an average serving of 450 grams per meal, we find a share of 6% waste per meal. This might appear very low, yet we know from interviews with the management that food waste is marginal due to high efficiency in preparation and little waste by students complemented by saving leftovers with TooGoodToGo and employee meals. Hence, we apply the 6% and split it equally between preparation waste and plate waste which results in 3% of the food waste for each. For preparation waste, we apply this percentage for raw ingredients that need to be transformed in some way such as peeled or trimmed or meat that needs to be cut whereas we do not apply it to pre-made ingredients such as tortellini. Management of the restaurant stated that plate waste of meat was marginal. Therefore, we assume the plate waste of meat and corresponding replacements in vegetarian meals to be 0. Hence, a share of 3% waste per meal is applied for non-meat and non-meat-replacement ingredients. Increased weight due to water absorption during cooking is not taken into account due to the minor effect thereof. The resulting organic waste is modeled using standard Swiss biowaste treatment to reflect typical local municipal composting.

For wastewater, this LCA considers the water requiring treatment after being used for processes like washing vegetables, cooking with water, or cleaning dishes. It is assumed that the water consumed for one meal corresponds to the water that requires wastewater treatment. This corresponding volume is subsequently modeled using an average Swiss municipal wastewater treatment.

3.7.2 Packaging Waste

As indicated above, the Agribalyse database considers packaging and waste treatment for processed products [1]. Raw products such as onions do not include packaging treatment yet. The impact of packaging cradle-to-grave makes up 1-2% of the total impact (EF 3.1 single score), depending on the scenario considered. Therefore, this study looks at packaging and its waste separately for having a better insight and being able to compare different scenarios. Packaging waste from plastic (HDPE and LDPE), glass, and tinplate steel are considered. To have an idea of the total impact of packaging, a separate LCA for packaging for one ready-made meal with a cradle-to-grave approach was performed. Management of the canteen told us that they have a considerable share of ingredients which they buy unpackaged and transport in plastic boxes. For these unpackaged products, we assume transportation with an HDPE box which weighs 1 kg and assume that 25 kg can be transported per box. This corresponds to the information provided by the restaurant management. We further assume that a box can be used 50 times. Hence, a box can transport $50 \times 25 \text{ kg} = 1'250 \text{ kg}$ wherefore the box of 1 kg divided by 1'250 kg results in 0.0008 kg of box per 1 kg of product.

To model the life cycle of the reusable transportation box, it is assumed that the boxes are manufactured in Germany via injection molding. Delivery of the finished boxes from the German factory to Switzerland is modeled over a transport distance of 700 km. During the operational use-phase, we assume that each box is washed after every single trip. This cleaning process is parameterized at 1.5 liters of tap water and 5 grams of industrial soap per 1 kg of crate capacity, translating to a scaled allocation of 0.003 liters of water and 0.0002 grams of soap per gram of transported food. Transportation at the end-of-life is assumed to amount to 15 km.

To model primary packaging of the ingredients, mass proxy multipliers were applied based on industrial packaging. Lightweight LDPE film was allocated at 3% of the ingredient weight

for dry goods, dairy, and spices, including bread, eggs, butter, and seasonings. Tinplate steel was modeled at 12% of the ingredient weight for concentrated, wet pantry items like tomato paste, while packaging glass was assumed for dense liquid goods such as olive oil, scaled at 40% of the food weight. These specific percentages are utilized because they reflect the physical tare weight ratios of industrial food packaging relative to their contents. For instance, the 3% LDPE multiplier captures the thin, high-surface-area nature of plastic film, the 12% steel ratio mirrors the structural thickness required for metal food cans, and the 40% glass multiplier accounts for the significant weight of rigid glass containers compared to lighter alternatives.



Figure 2: Transportation Box for Unpackaged Products.

Table 2 below shows the mass used for each type of packaging and the material share, which reflects the grams of packaging material required per gram of ingredient:

Packaging Type	Mass (g)	Material Share (g/g)
Transportation box (HDPE)	0.386	0.0008
Plastic packaging (LDPE)	3.74	0.0300
Glass jar	10.8	0.4000
Tinplate steel can	0.72	0.1200

Table 2: Packaging mass allocation and material intensity per Shakshuka meal.

The analysis for packaging is based on the Mediterranean Shakshuka menu. 3 distinct scenarios are evaluated: an incineration scenario, a recycling scenario, and a scenario that assumes the use of recycled materials in addition to recycling at the end-of-life. Recycled materials are already accounted for in both generic glass and tinplate steel, as these packaging types inherently incorporate high shares of recycled content. For recycling, the cut-off system model is applied, meaning the environmental burdens of the recycling process are allocated to the subsequent user of the material.

4 Life Cycle Impact Assessment Results

4.1 Comparison of meat-based and vegetarian meals impact

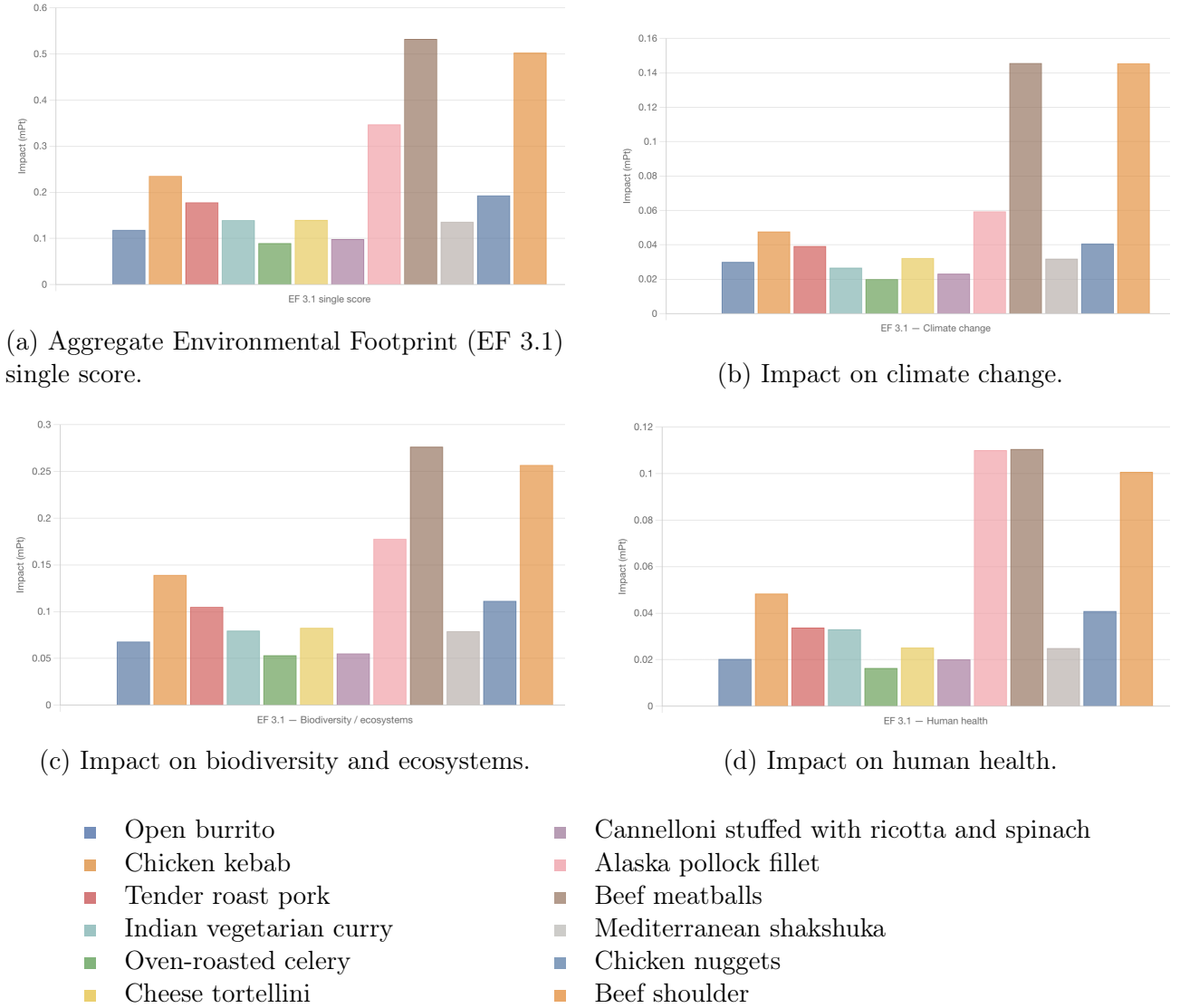


Figure 3: Multi-indicator environmental footprint analysis of twelve meals. Individual panels present results for the aggregate environmental footprint then for climate change, biodiversity, and human health, respectively.

Remark In the EF 3.1 methodology, mPt stands for milli-points (where 1 point = 1,000 mPt). An important relation to note is that **1 mPt corresponds to 35,9 kgCO₂-eq** [8]. In the A.2 , a table has been included to show the 1 mPt equivalents with applications to Switzerland.

The vegetarian meals Oven-roasted celery is the absolute lowest-impact meal on the entire chart, sitting well below 0.1 mPt. Coincidentally, it is the only vegan (fully plant-based) meal of the 6 vegetarian meals. Almost all the other vegetarian options (Cannelloni, Open burrito, Mediterranean shakshuka, Cheese tortellini, and Indian vegetarian curry) hover tightly in the 0.10 to 0.14 mPt range. The *key takeaway* is that when choosing a vegetarian meal, the environmental footprint is incredibly predictable and consistently low, usually staying under 0.15 mPt.

The red and white meat and fish meals This is where the numbers spike dramatically, but the type of animal protein matters immensely. Beef meatballs (highest overall at over 0.53 mPt) and Beef shoulder (just a fraction behind at 0.50 mPt) completely dominate the chart. Their impact is staggering, one single beef meatball meal has the same environmental footprint as roughly five burritos or six plates of roasted celery. This is likely driven by methane emissions and the heavy land/water use required to raise cattle. Alaska pollock fillet (fish) sits high at around 0.35 mPt, due to the import emissions, followed by Chicken kebab (0.24 mPt) and Tender roast pork (0.18 mPt)

The Key Takeaway is that beef exceeds all other types of meat in terms of the aggregate impact (single score) impact, while fish is also highly impactful (especially Alaska Pollock that travels from far).

4.2 Results by impact category

4.2.1 Climate change

This graph looks almost identical to the aggregate score, with beef completely dominating. Beef meatballs and Beef shoulder spike massively (0.145 mPt). Alaska pollock, which was quite high in the single score, drops significantly here, sitting only slightly above the Chicken kebab.

Hypotheses

- Methane: Cattle are ruminants, meaning they produce massive amounts of methane (CH_4) via enteric fermentation. Because methane is a highly potent greenhouse gas, beef's carbon footprint outpaces everything else by a mile.
- Pollock's low carbon profile: Wild-caught marine fish like Alaska Pollock don't require land clearing or feed cultivation. Their carbon emissions come almost entirely from the fuel burned by fishing trawlers, explaining why its climate impact is relatively modest compared to its overall single score.

4.2.2 Biodiversity and ecosystems

This is where the data gets interesting because the playing field flattens out with unexpected shifts. While beef is still the highest, it doesn't tower over the others nearly as violently. The Alaska Pollock (pink bar) and Chicken kebab (orange bar) creep much closer to the beef bars than in any other category.

Hypotheses

- Overfishing and marine: These suddenly skyrocket because the EF 3.1 Methodology includes marine eutrophication and ecotoxicity. Commercial trawling operations disrupt marine food webs, destroy seafloor habitats, and cause bycatch issues, which directly translate to heavy biodiversity degradation points.
- Pollock's low carbon profile: Wild-caught marine fish like Alaska pollock don't require land clearing or feed cultivation. Their carbon emissions come almost entirely from the fuel burned by fishing trawlers, explaining why its climate impact is relatively modest compared to its overall single score.

4.2.3 Human health

This graph features the most dramatic shift of all: Alaska pollock ties with Beef meatballs for the absolute worst spot on the chart (0.11 mPt), even beating out beef shoulder. The pink bar (pollock) shoots up to the maximum height. Meanwhile, the vegetarian options remain incredibly low, with Oven-roasted celery maintaining its position as the healthiest for the planet.

Hypotheses

- **Heavy fuel oil:** Marine fishing vessels typically burn heavy fuel oil (HFO) or marine diesel. The combustion of these low-grade fuels releases massive amounts of sulfur oxides (SO_x), nitrogen oxides (NO_x), and fine particulate matter ($\text{PM}_{2.5}$). As we noted in the EF 3.1 definitions, particulate matter inhalation is heavily weighted under human health due to respiratory illnesses.
- **Ammonia from livestock waste:** The beef and chicken scores here are likely driven by ammonia (NH_3) emissions from manure management and fertilizer application. Ammonia reacts in the atmosphere to form secondary particulate matter, creating a secondary driver for respiratory health impacts.

Conclusion on impact. If your only concern is climate change, avoiding beef is the single most effective action. However, if you care about biodiversity and human health, choosing wild-caught fish (like pollock) isn't the solution people think it is; the shipping emissions and marine disruption make it just as damaging as eating beef meatballs. Vegetarian meals remain the only choice that wins across all three dimensions simultaneously.

4.3 Breakdown by life cycle stage

The results reveal a stark asymmetry in where environmental pressures come from. As illustrated by the representative example of the Mediterranean shakshuka in Figure 6, agricultural production and raw material sourcing carry the vast majority of the environmental burden, while downstream phases play a comparatively negligible role. Crucially, this distribution pattern remains highly consistent across all twelve evaluated meals; regardless of the specific dish, the ingredients stage invariably dominates the life cycle profile, while transport, consumption, and waste consistently occupy minor shares of the total footprint.

Agricultural sourcing and ingredients The cultivation and production of raw ingredients constitute the dominant contributor, accounting for approximately 80% of the total Environmental Footprint (EF 3.1) single score. This impact is heavily tied to the agricultural phase of its primary components, such as tomatoes, peppers, onions, and eggs. This stage encompasses high-impact activities including synthetic fertilizer application, field machinery fuel consumption, and intensive water irrigation, demonstrating that the environmental destiny of the meal is largely decided before it ever leaves the farm gate.

Logistics and transportation In contrast to the emphasis often placed on "food miles" in public discourse, transport accounts for only about 10% of the total environmental impact. This indicates that the logistical supply chain required to move ingredients to processing facilities or distribution centers represents a minor fraction of the meal's life-cycle burden compared to the raw intensity of farming.

The consumption phase, primarily representing the energy required for refrigeration and cooking (e.g., gas or electric stove-tops), contributes roughly 10% to the overall score. This minimal footprint is driven by the efficiency gains inherent to canteen-scale preparation, where cooking in large batches substantially reduces per-meal energy demands compared to individual home cooking. This relatively small share is also context-dependent, as the analysis assumes the Swiss electricity mix, which is comparatively low-carbon. In regions relying more heavily on coal-based electricity, the contribution of the cooking and refrigeration stage would likely be higher. Nevertheless, within the Swiss context, consumer-side efficiency interventions, such as shifting cooking appliances or techniques, can only yield marginal environmental improvements compared to changes in ingredient sourcing.

Waste processing The waste stage is a negligible share of the stages. This indicates that post-consumer food waste emissions are mathematically dwarfed by the upstream farming phase, or that processing waste metrics have been integrated directly into the inventory of the ingredient life-cycle stage itself.

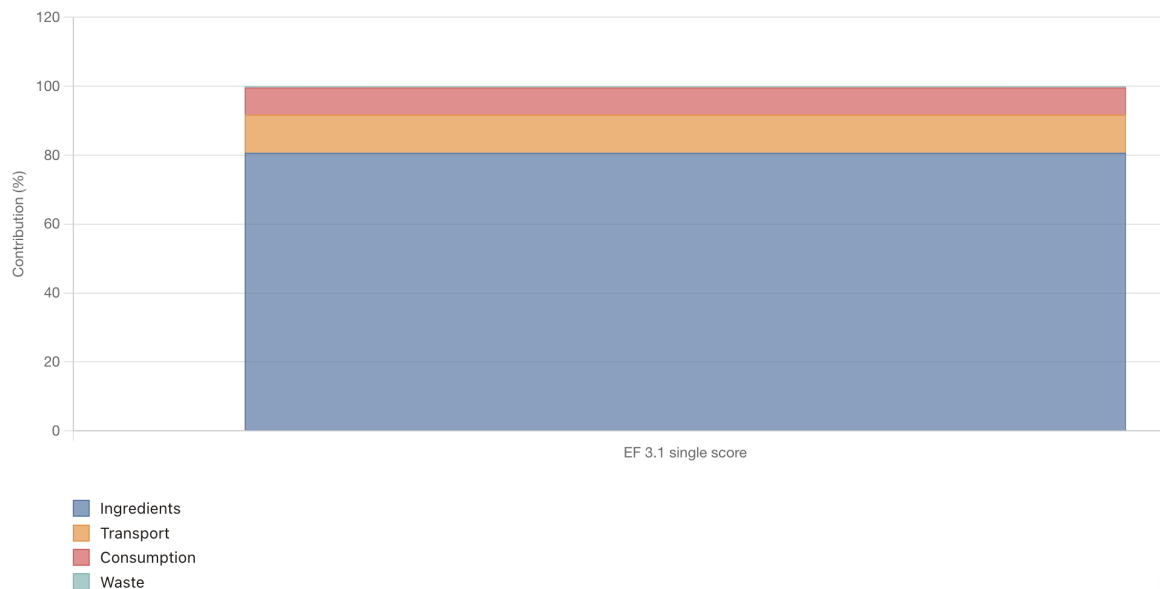
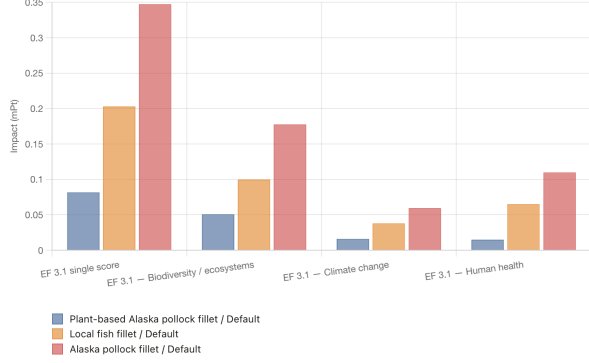


Figure 4: Life cycle stage breakdown and percentage contribution to the aggregate EF 3.1 single score for Mediterranean Shakshuka.

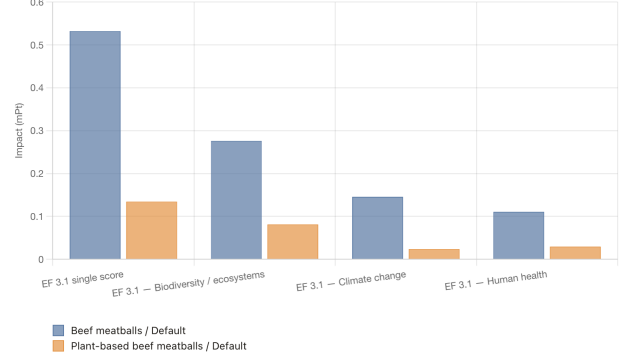
5 Reduction scenarios (Sensitivity analyses)

Following the identification of hot-spots, we considered several scenarios.

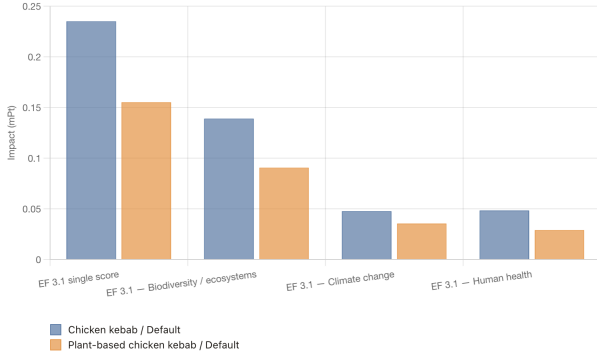
5.1 Scenario 1: Lowering the impact of non-vegetarian meals.



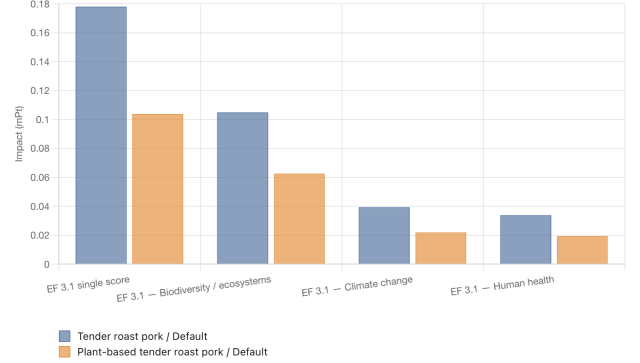
(a) Fish comparison with plant-based and local fish alternatives.



(b) Beef comparison with plant-based alternative.



(c) Chicken comparison with plant-based alternative.



(d) Pork comparison with plant-based alternative.

Figure 5: Comparison of non-vegetarian meals and alternative scenarios.

5.1.1 Modeling

The plant-based substitution scenario was modelled by replacing only the animal protein component with the closest available plant-based dataset in Agribalyse, using the same mass. The rest of the recipe system was kept unchanged, including side ingredients, transport, packaging, waste, and cooking processes.

Since no exact plant-based chicken, beef, pork, or fish datasets were available, plant-based patty or steak datasets were used as proxies for the animal-based dishes. These datasets remain relevant because they represent processed plant-based protein products made from comparable plant-derived ingredients, mainly wheat, soybeans, and legumes, and involve similar industrial transformation processes. The results should therefore be interpreted as a sensitivity analysis of shifting from animal-based to plant-based proteins, rather than as exact commercial product comparisons.

For the fish meal, two alternatives were tested. The first replaces the Alaska Pollock fillet with a plant-based fish alternative. The second replaces it with a more local European Demersal fish option, used as a proxy for a locally sourced white fish. This allows us to distinguish between the effect of changing protein type and the effect of changing sourcing location.

5.1.2 Results

Figure 5 presents the environmental impacts of the non-vegetarian meals and their corresponding plant-based or local alternatives across the EF 3.1 single score, biodiversity, climate change, and human health indicators.

Overall, plant-based substitutions reduce impacts across all categories, but the size of the reduction depends strongly on the original protein. The strongest reduction is observed for beef meatballs, where the single score decreases by around 75%, from 0.53 mPt to 0.13 mPt. This confirms that beef is the main hot spot and that replacing it offers the largest environmental benefit.

The fish scenario also shows a strong reduction potential. Replacing Alaska Pollock with a plant-based alternative reduces the single score by around 76%, from 0.35 mPt to 0.08 mPt. The local European fish option also improves the result, but less strongly, with a reduction of around 42%, reaching 0.20 mPt. This suggests that local sourcing helps, but changing the protein type has a larger effect than changing the fish origin alone.

For chicken and pork, the reductions are positive but more moderate. The chicken kebab decreases by around 34%, from 0.24 mPt to 0.16 mPt, while tender roast pork decreases by around 42%, from 0.18 mPt to 0.10 mPt.

These results show that plant-based substitutions are beneficial across all non-vegetarian meals, with the largest gains achieved by replacing beef and Alaska Pollock.

5.2 Scenario 2: Comparing meals by Daily Functional Unit (DFU).

For the DFU comparison, six representative meals were selected from the menu portfolio: four non-vegetarian dishes, one vegetarian dish, and one vegan dish. This selection reflects the main dietary categories in the baseline menu and allows meals with different ingredient compositions and nutritional profiles to be compared. The four non-vegetarian dishes were included because meat and fish were identified as important contributors to the overall climate impact, while the vegetarian and vegan meals serve as plant-based reference points.

The Daily Food Unit (DFU) is Eaternity's standardized functional unit for comparing the environmental impact of meals on a nutritional basis. Instead of comparing meals only by weight, such as per 100 g or per portion, the DFU relates the environmental impact to the nutritional contribution of the meal. The calculation considers five dimensions: protein, fat, energy, water, and dry weight.

This approach is important because meals can differ strongly in nutritional density. A meal with low absolute emissions may still have a higher CO₂eq/DFU value if it provides relatively little nutritional value. Therefore, the DFU allows meals to be compared not only by their total greenhouse gas emissions, but also by their climate efficiency in relation to the nutrition they provide.

5.2.1 Calculation Methodology

The Daily Food Unit (DFU) is calculated as the average contribution of a meal across five nutritional dimensions: protein, fat, energy, water, and dry weight. Following Eaternity’s methodology, the DFU is defined as:

$$DFU = \frac{R_{\text{protein}} + R_{\text{fat}} + R_{\text{energy}} + R_{\text{water}} + R_{\text{dry}}}{5} \quad (2)$$

where the five nutritional ratios are defined as follows:

$$R_{\text{protein}} = \frac{g_{\text{protein}}}{50} \quad (3)$$

$$R_{\text{fat}} = \frac{g_{\text{fat}}}{66} \quad (4)$$

$$R_{\text{energy}} = \frac{kJ_{\text{energy}} - (17g_{\text{protein}}) - (37g_{\text{fat}})}{6000} \quad (5)$$

$$R_{\text{water}} = \frac{g_{\text{water}}}{2500} \quad (6)$$

$$R_{\text{dry}} = \frac{g_{\text{dry weight}}}{600} \quad (7)$$

5.2.2 Example Calculation: Chicken Kebab

As an example, the DFU calculation is illustrated using one of Alpine’s most sold menus, the Chicken Kebab.

Dimension	Input value	Formula applied	Ratio
Protein	40 g	40/50	0.80
Fat	44 g	44/66	0.66
Energy	4079 kJ	$\frac{4079 - (17 \times 40) - (37 \times 44)}{6000}$	0.29
Water	544 g	544/2500	0.21
Dry weight	193 g	193/600	0.32
Final DFU	–	Average of five ratios	0.46

Table 3: DFU calculation for the Chicken Kebab menu

This menu has a serving size of approximately 700 g and a total environmental impact of 0.0478 mPt, calculated using the SimpleLCA tool. Following the conversion method described in Chapter 5.1, the mPt result is converted into absolute greenhouse-gas emissions in grams of CO₂ equivalents:

$$0.0478 \times 35.9 \times 1000 = 1716 \text{ g CO}_2\text{eq} \quad (8)$$

The nutritional information required for the DFU calculation was obtained from the Ciquál food composition database. For the Chicken Kebab menu, the nutritional profile used was 40 g protein, 44 g fat, 4079 kJ energy, 544 g water, and 193 g dry weight. Using these values, the five nutritional ratios were calculated as shown in Table 3, leading to a final DFU of 0.46.

This means that the Chicken Kebab menu covers approximately 46% of the average daily nutritional requirement across the five DFU dimensions.

Finally, the normalized climate impact is obtained by dividing the meal’s absolute greenhouse-gas emissions by its DFU:

$$\frac{1716 \text{ g CO}_2\text{eq}}{0.46 \text{ DFU}} = 3730 \text{ g CO}_2\text{eq/DFU} \quad (9)$$

Thus, the Chicken Kebab menu has a normalized climate impact of approximately 3730 g CO₂eq per DFU.

5.2.3 Climate Rating Context

By standardizing the impact to 3,730 g CO₂eq per DFU, we can directly evaluate the Chicken Kebab using Eaternity’s global rating system:

Rating	CO ₂ eq/DFU Threshold	Improvement %	Meaning
A	< 1,947 g	≥ 50% better	Climate friendly
B	< 3,894 g	> 0% better	Better than average
C	< 7,787 g	> −100%	Up to 2× the average
D	< 15,574 g	> −300%	Up to 4× the average
E	> 15,574 g	≤ −300%	More than 4× the average

Table 4: Eaternity Global Climate Rating System

The Chicken Kebab scores just under the threshold, earning it a **B rating (Better than average)**.

5.2.4 Observations from the baseline meals

The baseline results show that absolute environmental impact and DFU-normalized climate impact can lead to different interpretations. For example, the Oven-roasted Celery has one of the lowest absolute impacts in the SimpleLCA results, but it receives a C rating after DFU normalization. This is due to its relatively low nutritional density: although the dish emits little CO₂eq in total, it also contributes a smaller share of the daily nutritional requirement. Therefore, its emissions per DFU are higher than its absolute impact alone would suggest.

This highlights the importance of the DFU approach, as it evaluates climate impact in relation to the nutritional value provided by each meal. The Beef Meatballs remain the most climate-intensive baseline meal, with 11,922.54 g CO₂eq/DFU and a D rating. In contrast, the Mediterranean Shakshuka performs well, achieving a B rating because of its comparatively low emissions and higher DFU value.

Meal	CO ₂ eq/meal (kg)	Final DFU	CO ₂ eq/DFU (g)	Rating
Chicken Kebab	1.716	0.46	3,722.20	B
Alaska Pollock Fillet	2.132	0.32	6,656.39	C
Tender Roast Pork	1.411	0.34	4,103.05	C
Oven-roasted Celery	0.718	0.15	4,900.27	C
Mediterranean Shakshuka	0.872	0.44	1,998.43	B
Beef Meatballs	5.241	0.44	11,922.54	D

Table 5: Baseline Meals: Absolute and DFU-Normalized Climate Impact

5.2.5 Baseline meals and their vegetarian alternatives

To evaluate the potential for climate mitigation through dietary shifts, the original six baseline meals were compared with their vegetarian or plant-based alternatives. This comparison shows how replacing meat or fish components affects the climate impact of a meal. Using the DFU ensures that the meals are compared not only by their emissions, but also by the nutritional value they provide.

Table 6 displays the adjusted menus where meat and fish components were replaced with vegetarian alternatives. The Oven-roasted celery and Mediterranean shakshuka are omitted from this table as they already serve as meatless baselines.

Meal Alternative	CO ₂ eq/meal (kg)	Final DFU	CO ₂ eq/DFU (g)	Rating
Chicken Kebab Veggie	1.274	0.46	2,744.33	B
Alaska Pollock Fillet Veggie	1.585	0.33	1,790.58	A
Tender Roast Pork Veggie	0.786	0.30	2,591.22	B
Beef Meatballs Veggie	0.862	0.39	2,194.16	B

Table 6: Vegetarian/Plant-Based Alternatives: Absolute and DFU-Normalized Climate Impact

5.2.6 Analysis: The Impact of Dietary Shifts

By comparing the baseline meat and fish dishes to their vegetarian alternatives, we observe a clear reduction in CO₂eq per DFU across all four substituted meals. This means that the vegetarian alternatives provide the same nutritional function with a lower climate impact. Table 7 quantifies these improvements and shows how the veggie alternatives affect the Eaternity ratings.

Menu Item	Baseline (g)	Veggie Alternative (g)	Reduction	Rating Shift
Chicken Kebab	3,722	2,744	−26%	B → B
Alaska Pollock	6,656	1,791	−73%	C → A
Tender Roast Pork	4,103	2,591	−37%	C → B
Beef Meatballs	11,923	2,194	−82%	D → B

Table 7: Direct Impact Analysis: Baseline vs. Vegetarian Alternatives

Key observations DFU normalization changes the interpretation of some meals. A low absolute CO₂eq impact does not automatically imply a good climate rating. For example, Oven-roasted Celery has the lowest absolute impact among the baseline meals, but still receives a C rating because its nutritional density is relatively low, which increases its impact per DFU.

Beef remains the dominant hotspot. Beef Meatballs have the highest CO₂eq/DFU value among the baseline meals, with 11,922.54 g CO₂eq/DFU and a D rating. After replacing the beef component with a vegetarian alternative, the value drops to 2,194.16 g CO₂eq/DFU and the rating improves to B.

All vegetarian alternatives reduce the normalized climate impact of the meal. The strongest improvements are observed for Beef Meatballs and Alaska Pollock Fillet, with reductions of 82% and 73%, respectively. Chicken Kebab shows a smaller reduction of 26%, but still improves even though its rating remains unchanged.

Overall, the DFU-based analysis confirms that substituting meat and fish with plant-based alternatives can substantially improve the climate efficiency of a meal while maintaining its nutritional function.

5.3 Scenario 3: Types of cutlery

To evaluate the environmental trade-offs between different dining setups, this scenario models three distinct cutlery and dishware systems used at Foodlab Alpine. To isolate the material and life-cycle impacts, the analysis relies on a set of core operational parameters regarding how each system is handled, washed, and maintained.

Parameter	Operational framework
Ceramic (On-site)	Assumes heavy-duty, reusable ceramic plates, washed on-site using standard industrial dishwashers.
Plastic (On-site)	Models reusable, lightweight plastic dishware washed on-site, mirroring a fast-casual or high-volume cafeteria setup.
reCIRCLE (Takeaway)	Represents a closed-loop takeaway system where high-grade polymer boxes are reused across a managed network, assuming regional Swiss collection and washing.
Washing baseline	All on-site washing scenarios use the local Swiss electricity and water mix, reflecting centralized, high-efficiency machine cleaning.

Table 8: Operational assumptions for the three cutlery systems.

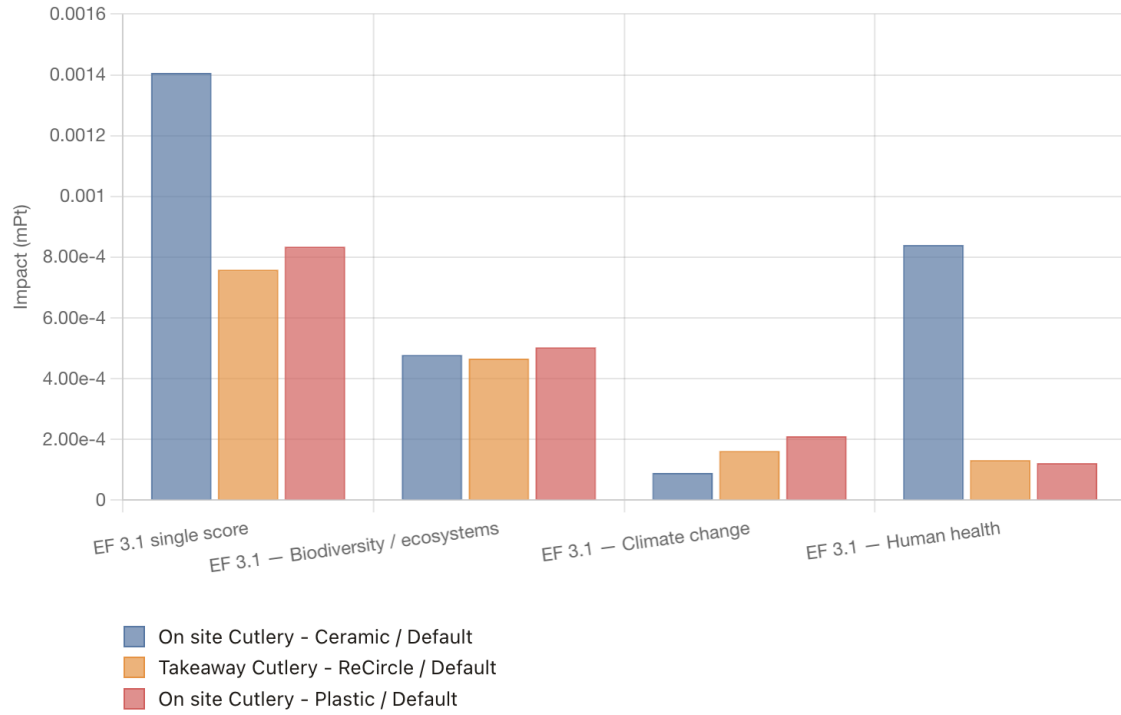


Figure 6: Comparison of life cycle environmental impacts of types of cutlery.

This scenario reveals a clear tension between carbon emissions and human toxicity across the 3 cutlery systems. On-site ceramic cutlery achieves the lowest carbon footprint (Climate change) because its exceptional long-term durability dilutes its raw material impacts over hundreds of uses; however, the heavy industrial mining and high-temperature kiln firing required to make ceramics release significant inorganic particulates, giving it the worst Human health score and highest overall single score. On the contrary, on-site plastic cutlery is chemically cleaner to manufacture (low human health impact), but incurs the highest climate penalty due to its fossil-fuel origins and the direct greenhouse gasses released if it is incinerated at the end of its life.

Interestingly enough, the best environmental balance is the reCIRCLE takeaway cutlery. By using local, high-grade polymers optimized for a closed-loop Swiss recycling network, it minimizes resource waste and ecosystem disruption (Biodiversity), ultimately securing the lowest overall environmental single score. However, it would be impractical to use it indoors. If we forced everyone to use a reCIRCLE box for sit-down dining, we would be replacing a durable material with a plastic-based compound that scratches more easily under heavy, constant daily knife use, meaning it would have to be retired sooner than a ceramic plate.

5.4 Scenario 4: Types of Packaging

For packaging, we look at the 3 scenarios of incineration at the end-of-life, recycling at the end-of-life, and using recycled materials in addition to recycling at the end-of-life.

Note: The appendix A.4 shows more detailed graphs for the impact on human health, climate change, and biodiversity/ecosystems.

5.4.1 Incineration Scenario

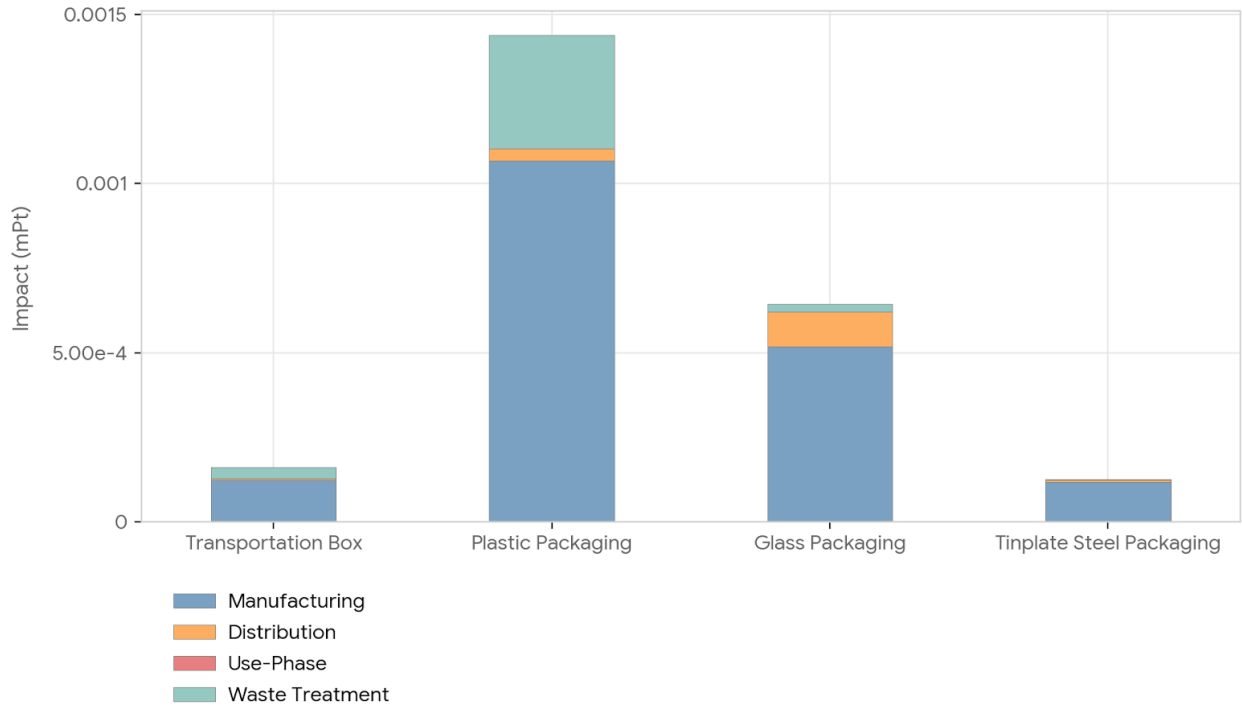


Figure 7: Incineration Scenario: EF 3.1 single score

The figure above illustrates the environmental impact of the four packaging types considered for the Mediterranean Shakshuka meal under the baseline incineration scenario. Plastic packaging demonstrates the highest absolute impact, with the manufacturing stage acting as the dominant driver across all packaging materials.

As detailed in the table below, the impact per gram of packaging material is highest for the HDPE transportation box and lowest for the glass jar. However, due to the extremely low material intensity required per ingredient for the transportation box, it remains a highly efficient packaging form per serving. Conversely, while glass has a low environmental impact per gram, the high mass required per serving makes it significantly less efficient overall.

Table 9: Packaging mass and environmental impact per Shakshuka meal.

Packaging Type	Mass (g)	Abs. Impact (mPt)	Impact/g (mPt/g)
Transportation box (HDPE)	0.386	1.622e-04	4.202e-04
Plastic packaging (LDPE)	3.74	1.439e-03	3.848e-04
Glass jar	10.8	6.448e-04	5.970e-05
Tinplate steel can	0.72	1.253e-04	1.740e-04

5.4.2 Recycling Scenario

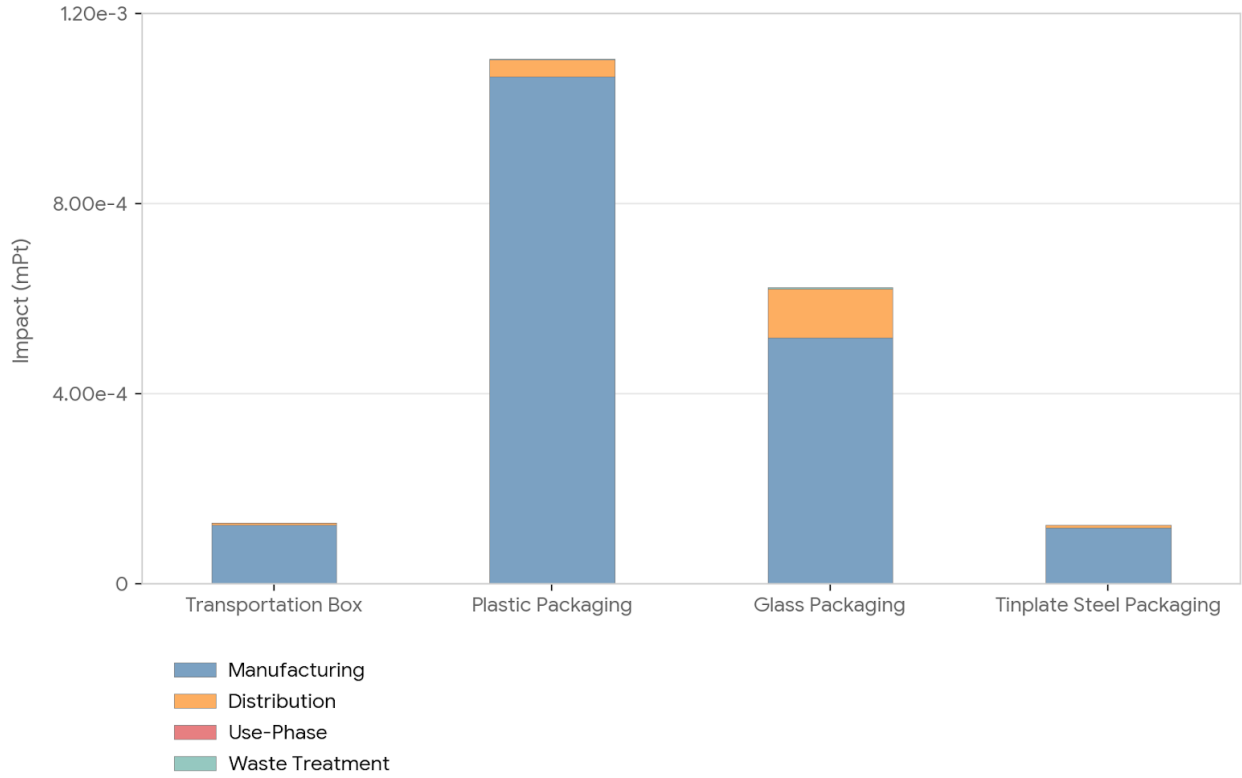


Figure 8: Recycling Scenario: EF 3.1 single score

Shifting the end-of-life treatment from incineration to recycling yields the most substantial benefits for polymer-based packaging due to avoiding the release of fossil carbon stored in the plastics during combustion. LDPE plastic packaging sees a 23% decrease in the EF 3.1 single score compared to the baseline, while the HDPE transportation box impact decreases by 21%.

In contrast, the recycling scenario offers only marginal improvements for glass and tinplate steel, reducing their overall impacts by just 3% and 1%, respectively. This highlights that diverting plastics from incineration avoids significant emissions associated with the thermal treatment of fossil-based materials, effectively neutralizing the waste treatment burden visible in the baseline scenario.

Table 10: Recycling Scenario: Packaging mass and environmental impact per Shakshuka meal.

Packaging Type	Mass (g)	Abs. Impact (mPt)	Impact/g (mPt/g)
Transportation box (HDPE)	0.386	1.275e-04	3.303e-04
Plastic packaging (LDPE)	3.74	1.104e-03	2.952e-04
Glass jar	10.8	6.228e-04	5.767e-05
Tinplate steel can	0.72	1.240e-04	1.722e-04

5.4.3 Recycled Materials Scenario

The third scenario evaluates the integration of recycled materials into the manufacturing phase alongside end-of-life recycling. This combined approach leads to drastic environmental savings for the plastic packaging categories by effectively shrinking the primary manufacturing burden.

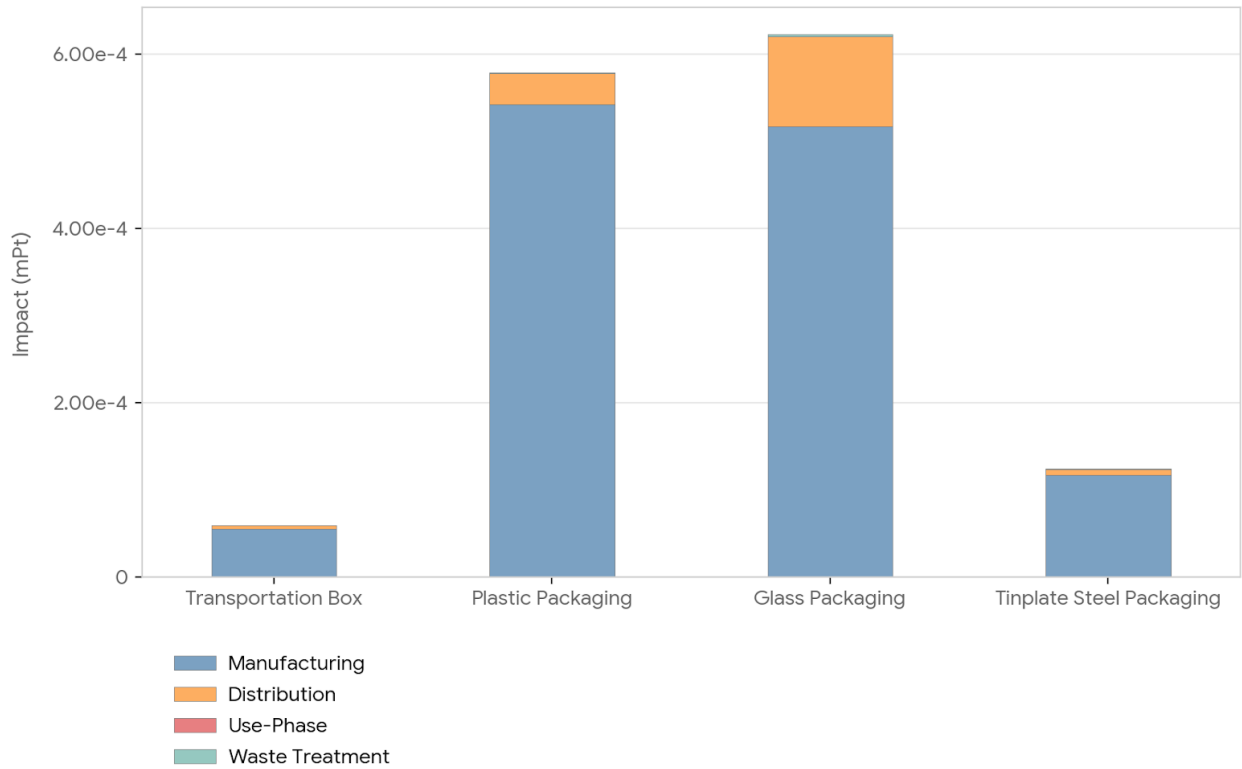


Figure 9: Recycled Materials: EF 3.1 single score

The impact of the HDPE transportation box drops to just 36% of its original baseline, while LDPE plastic packaging falls to 40% of its baseline. The glass and tinplate steel categories do not reflect further improvements in this specific modeling scenario, remaining at 97% and 99% of their baseline impacts. This limited improvement is because these materials already assume a high industry-average share of recycled content in the baseline scenario, leaving significantly less potential for theoretical gains compared to the more virgin-heavy plastic streams. These results emphasize that substituting virgin raw materials with recycled content is the most critical lever for mitigating the overall life cycle impact of plastics.

Table 11: Recycled Materials Scenario: Packaging mass and environmental impact per Shakshuka meal.

Packaging Type	Mass (g)	Abs. Impact (mPt)	Impact/g (mPt/g)
Transportation box (HDPE)	0.386	5.892e-05	1.526e-04
Plastic packaging (LDPE)	3.74	5.790e-04	1.548e-04
Glass jar	10.8	6.228e-04	5.767e-05
Tinplate steel can	0.72	1.240e-04	1.722e-04

5.4.4 Comparison and Key Observations

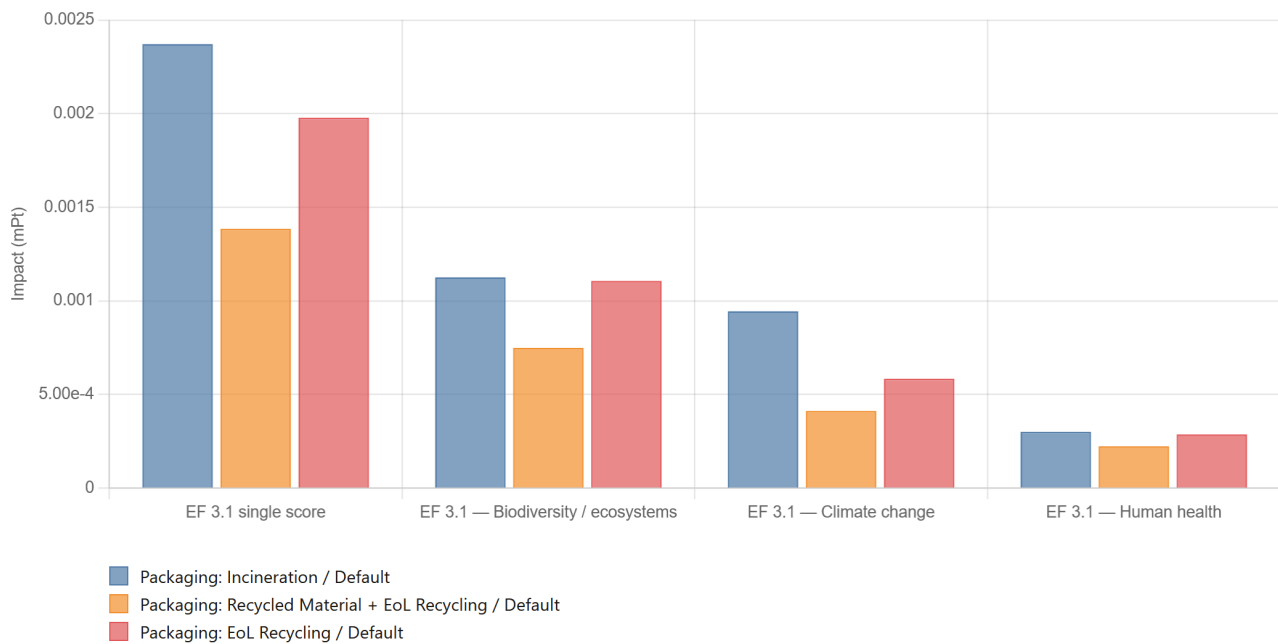


Figure 10: Aggregated impact comparison of packaging scenarios across EF 3.1 indicators

The aggregated environmental performance of all packaging materials combined per Shakshuka meal is summarized in Figure 10. Across all four EF 3.1 impact indicators, the baseline incineration scenario consistently dictates the highest environmental burden.

Introducing end-of-life recycling provides a moderate reduction in the total environmental impact. As established in the individual material analysis, this benefit is almost exclusively driven by diverting polymer-based packaging (LDPE and HDPE) from thermal treatment, thereby eliminating the associated incineration emissions. However, because the primary manufacturing burden remains unchanged, the overall system reduction is limited. Materials like glass and tinsplate steel, which possess high baseline manufacturing impacts but less for incineration, remain largely unaffected in this secondary scenario.

The most substantial environmental savings are achieved under the third scenario, which integrates recycled materials into the manufacturing phase alongside end-of-life recycling. By substituting virgin materials, the dominant manufacturing impacts of plastic packaging are drastically reduced. The aggregated data shows that this approach nearly halves the total packaging footprint for the single score, climate change, and human health indicators compared to the baseline.

A defining observation from this comparison is that while end-of-life waste management is beneficial, the primary material production phase ultimately dictates the vast majority of the packaging footprint. For high-mass items like the glass jar, shifting waste treatment scenarios offers negligible life cycle benefits per serving. Therefore, the most effective strategy for mitigating packaging impacts is twofold: prioritizing material lightweighting - as shown by the highly efficient, low-mass HDPE transportation box - and maximizing the proportion of recycled content in polymer-based materials to suppress manufacturing burdens.

5.5 Scenario 5: Macro-level - Implementation of more vegetarian days

To evaluate the reduction potential for a food caterer, this section models a series of macro-level menu modifications against the baseline operations of Foodlab Alpine. Using consumption data from a representative semester week (February 23–27, 2026) [12], we assessed the ecological implications of increasing the frequency of vegetarian days. The analysis progresses from targeted single-meal substitutions (red meat vs. white meat vs. fish) to a full-vegetarian weekly menu.

Crucially, this model operates under the strong assumption that the total number of meals served remains unchanged. In reality, this is a major variable: on one hand, fewer students might choose to eat if meat isn't an option; on the other hand, plant-based meals could be priced cheaper, which might actually drive up demand. Since we don't know which of these two effects would dominate, keeping the volume constant gives us a clean baseline. The full set of assumptions is shown in the table below.

Assumption	Definition & rationale
Constant volume	Total weekly meal counts remain strictly unchanged. While meat removal might deter some diners and lower prices might attract others, holding volume constant isolates the impact of ingredient sourcing alone.
Component matching	The structural design of the menus remains identical to the status quo, swapping meat components exclusively for functionally equivalent plant-based alternatives.

Table 12: Core macro-level modeling assumptions.

We first modeled the impact of adding just one extra veggie day to the weekly menu in addition to the existing veggie day, specifically testing what happens when we replace either the pork, chicken, beef, or fish meal of the week. After analyzing those individual shifts, we then modeled a comprehensive "Full Veggie Week" scenario. This analysis used data provided by Foodlab Alpine on the number of meals sold during a typical semester week (February 23–27, 2026), including the share of vegetarian meals and non-vegetarian meals. For the full veggie week, the menu items remain identical but use plant-based meat substitutes, meaning students wouldn't necessarily notice a change. The detailed table of results is included in the Appendix A.3.

Scenario	Total Impact	Relative Savings
Status quo week	487.20	Baseline
2 Veggie Days - no pork	466.18	4%
2 Veggie Days - no fish	422.26	17%
2 Veggie Days - no beef	380.13	22%
Vegetarian week	246.26	49%

Table 13: Summary of weekly total impacts and environmental savings across scenarios

The results show that the environmental payoff depends entirely on which protein you replace. Swapping out pork or chicken for an extra veggie day has a negligible impact, yielding just a 4 to 5% savings on the total ecological footprint. However, cutting out the menu's major emitters yields massive, consequential savings: replacing the fish meal brings a 17% reduction, while replacing the beef meal cuts the footprint by 21%. Moving forward, the SV Group's

highest-leverage strategy isn't to push broad vegetarian mandates, but to strictly limit beef and phase out fish transported from far away.

5.6 Comparison of scenario results

To evaluate the reduction potential effectively, the analysis is split into individual meal choices (what a customer selects or what goes into a single serving) and macro-level operations (weekly menu planning). All values are focused on absolute environmental savings (Δ mPt) or absolute climate impact savings, using the conversion metric $1 \text{ mPt} = 35.9 \text{ kg CO}_2\text{eq}$.

5.6.1 Individual meal choices: Micro-levers

This category compares the levers available per individual serving. We look at how shifting from animal-based proteins to plant-based proxies drastically alters a single dish's footprint compared to tweaking its operational packaging or cutlery.

Table 14: Comparison of micro-levers for individual meal choices

Scenario	decision	Baseline impact	Optimized impact	Environmental savings (Δ)	Climate savings (CO_2eq)
Beef to veggie		11,923 g / DFU	2,194 g / DFU	$\sim 0.271 \text{ mPt}$ (per DFU)	9.73 kg
Fish to veggie		6,656 g / DFU	1,791 g / DFU	$\sim 0.136 \text{ mPt}$ (per DFU)	4.87 kg
Pork to veggie		4,103 g / DFU	2,591 g / DFU	$\sim 0.042 \text{ mPt}$ (per DFU)	1.51 kg
Chicken to veggie		3,722 g / DFU	2,744 g / DFU	$\sim 0.027 \text{ mPt}$ (per DFU)	0.98 kg
On-site cutlery: Ceramic vs. plastic		$\sim 0.00140 \text{ mPt}$	$\sim 0.00083 \text{ mPt}$	0.00057 mPt	0.020 kg
Packaging: Plastic baseline vs. re-cycled		0.00144 mPt	0.00110 mPt	0.00034 mPt	0.012 kg

Moving down the protein hierarchy The data reveals a steep progression in savings as we move down the animal protein spectrum. Shifting away from chicken or pork provides modest improvements (1 to 1.5 kg CO_2eq saved) compared to stepping up to fish and ultimately beef. Swapping a single beef item for a veggie alternative saves 9.73 kg CO_2eq , roughly 10 times the savings of swapping a chicken item.

Remark: Crucially, the DFU metrics track this exact progression, proving that these shifts represent a genuine optimization of climate efficiency. The normalized impact per DFU drops drastically for the high-impact items—plummeting by 82% for beef and 73% for fish. This reveals the vegetarian alternatives provide the same core nutritional function at a fraction of the environmental cost.

Leveraging cutlery and packaging choices The analysis on cutlery shows a surprising paradox. Even though ceramic cutlery has a better climate change score due to reuse durability, its heavy mining and manufacturing give on-site plastic an better overall aggregate score.

Meanwhile, the reCIRCLE takeaway system beats both on-site plastic and ceramic in overall single-score impact by balancing low manufacturing stress with circular economy benefits. Regrettably, reCIRCLE is simply not adapted for on-site dining. In addition, optimizing plastic packaging loops to use recycled content yields a savings of 0.00034 mPt per meal. While a positive operational step, this savings is entirely negligible when compared to simply changing the recipe’s protein component.

5.6.2 Restaurant operations: Macro-levers

This scenario steps back from individual ingredients to look at Foodlab Alpine’s total footprint across a full week. It highlights how systemic menu architecture changes the restaurant’s total environmental volume. Optimizing Foodlab Alpine’s menu structure boils down to a clear hierarchy of leverage. Achieving a substantial $\sim 20\%$ operational footprint reduction does not require systemic mandates; it can be unlocked simply by replacing high-impact red meat or fish for one day a week, saving up to 3,843.8 kg CO₂eq. Conversely, targeting chicken or pork offers negligible environmental returns ($\sim 4\%$ savings) and is practically not worth the operational shift. While a full vegetarian week yields a 49% total reduction, this model operates under the fragile, ideal assumption that student volume remains completely static. In reality, this introduces a major behavioral risk: without careful pricing strategies or seamless recipe execution (ie. so that students barely spot the change!), a strict plant-based switch could drive students to eat elsewhere, displacing the environmental footprint rather than eliminating it.

6 Results in Context

6.1 From one lunch to food habits

The results show that meal choices at FoodLab Alpine are environmentally meaningful, even at the scale of a single lunch. In the baseline comparison, the lowest-impact meal, Oven-roasted Celery, remains below 0.1 mPt, whereas the highest-impact meal, Beef Meatballs, exceeds 0.5 mPt. Thus, one beef-based lunch can generate roughly five to six times the environmental burden of one of the lowest-impact vegetarian meals.

What makes this important is repetition. Since FoodLab Alpine serves several hundred meals per day and similar choices are made over weeks and semesters, even small differences per meal accumulate into substantial environmental effects. This is also reflected in the weekly scenario analysis, where a fully vegetarian week reduces the total weekly impact by about 49%.

6.2 What the comparison reveals

The meal comparison makes clear that the environmental burden is not distributed evenly across proteins. Beef is by far the dominant hotspot, while poultry and pork occupy an intermediate position. Vegetarian meals remain consistently lower-impact overall. At the same time, the results show that fish cannot automatically be treated as a low-impact alternative: Alaska Pollock performs less severely in climate terms than beef, but poorly in biodiversity and human health.

This means that the issue is not simply “meat versus vegetarian,” but the repeated inclusion of specific high-impact ingredients in everyday menus. The composition of the full dish still matters, especially once nutritional value is considered through the DFU-based analysis.

6.3 Why this matters in practice

The main implication of these results is not simply that vegetarian meals perform better on paper. The stakeholder discussions showed that vegetarian options face real barriers related to student preferences, perceived satiety, and cost. Many students prefer simple and filling comfort food, often meat-based, while high-quality vegetarian cuisine can be more expensive to prepare.

The challenge for FoodLab Alpine is therefore not only to offer lower-impact meals, but to make them attractive, satisfying, and affordable enough to be chosen in practice. In this sense, the strongest leverage lies less in small operational adjustments and more in menu design itself.

7 Recommendations and priorities for action

The first priority should be targeted menu substitution rather than broad vegetarian mandates. FoodLab Alpine should focus on replacing the highest-impact animal protein sources, especially beef and imported fish, while keeping the most popular meals, such as the chicken kebab, with lower-impact ingredients.

Second, plant-based meals should be improved through recipe design rather than sustainability labeling alone, since labeling a dish as sustainable can sometimes create the impression that it is less indulgent, less filling, or less appealing. The goal should be to ensure that lower-impact meals remain attractive, satisfying, and affordable, so that students choose them voluntarily rather than turning away from them.

Third, EPFL could introduce a second vegetarian day on campus. Since EPFL restaurants already operate one vegetarian day per week, adding a second day would be a realistic extension of the current system. However, this should be implemented gradually and monitored carefully, as forcing too rapid a shift could reduce restaurant attendance or displace consumption elsewhere.

Fourth, EPFL could test a semester-based vegetarian meal plan as a supporting incentive. Students could buy a fixed number of vegetarian meals in advance at a lower price than the usual on-the-spot price. This would reduce the price barrier, support demand stability for restaurant operators, and encourage repeated exposure to vegetarian meals.

Finally, higher-impact improvements should focus primarily on menu composition. Operational actions such as recycled packaging, cutlery choices, and waste reduction remain useful, but should be treated as secondary levers compared to changing the protein structure of meals.

8 Conclusion

The goal of this study was to evaluate the environmental impacts of meals served at EPFL’s FoodLab Alpine and provide an estimate of the potential benefits of shifting towards plant-based options and optimizing restaurant operations.

In the current situation, vegetarian meals clearly induce lower environmental impacts across all considered indicators, consistently generating a highly predictable and low environmental footprint under 0.15 mPt. In contrast, beef-based meals are the dominant environmental hotspot; a single Beef Meatballs meal generates the highest overall impact (0.53 mPt), which is equivalent to the footprint of roughly five vegetarian burritos. Across all evaluated menus, the agricultural cultivation and production of raw ingredients account for approximately 80% of the total environmental footprint.

In the future, replacing high-impact animal proteins with plant-based alternatives would enable a large reduction in current greenhouse gas emissions and overall environmental impacts. For example, replacing the beef component with a plant-based alternative reduces the single score environmental impact by 75% and lowers the DFU-normalized climate impact by 82%. At a macro-level, swapping out just the beef meals for a vegetarian option over a standard week cuts the restaurant’s total weekly environmental footprint by 22%. For packaging, integrating recycled materials into the manufacturing phase alongside end-of-life recycling nearly halves the total packaging footprint for plastic items.

However, selecting the most sustainable operational components is not always straightforward and depends on the type of materials, practical dine-in constraints, and end-of-life treatments. Should all proposed sustainability options be implemented, the choices might depend on the priority given to specific indicators.

Limitations of the study include the following:

- **Recipe reconstruction:** Exact recipes used by FoodLab Alpine are confidential, meaning meal compositions and precise ingredient ratios had to be approximated.
- **Data gaps and assumptions made:** Specific measurements for kitchen operations were unavailable. Consequently, energy use for active cooking, water consumption, and 48-hour cold storage were dynamically modeled using standardized SV Group KPIs rather than measured. Additionally, an assumed 6% food waste rate was applied based on literature.
- **Database matching:** Minor ingredients and packaging were represented by mass proxy multipliers. Furthermore, exact plant-based datasets for chicken, beef, pork, or fish were unavailable in the Agribalyse database; therefore, generic plant-based patty or steak datasets were utilized as proxies.
- **Limits to comparability:** Because the comparative assessment relies on reconstructed recipes and proxy datasets, this indirect comparison introduces a potential limitation. The results should be interpreted as a sensitivity analysis of shifting from animal-based to plant-based proteins, rather than an exact commercial product comparison.

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A Appendix

A.1 Supplier Transportation Distance Calculation

This appendix presents the detailed calculation of the weighted average supplier transportation distance (d_{avg}) used in the LCA.

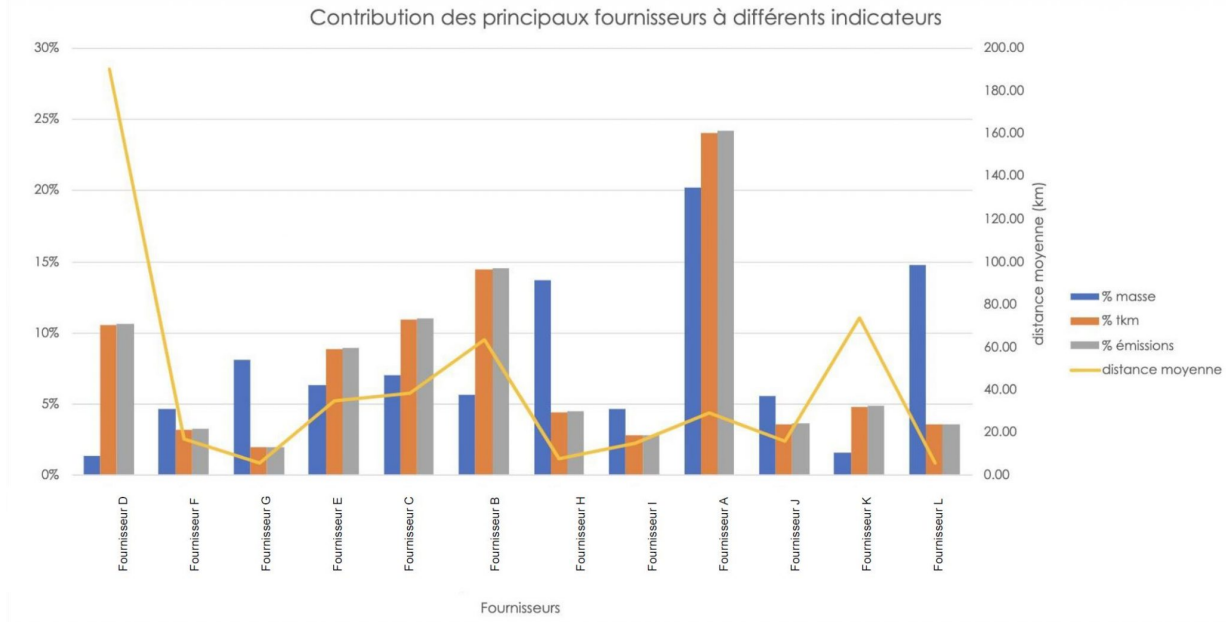


Figure 11: Contribution of major suppliers to transported mass, tonne-kilometres, emissions, and average delivery distance [6].

The weighted average supplier distance was calculated as:

$$d_{avg} = \frac{\sum_i (\%Mass_i \times Distance_i)}{\sum_i \%Mass_i}$$

Detailed calculation:

$$\begin{aligned}
 \sum_i (\%Mass_i \times Distance_i) &= (1.3 \times 190) + (4.6 \times 18) + (8.0 \times 8) \\
 &\quad + (6.3 \times 35) + (7.0 \times 39) + (5.6 \times 65) \\
 &\quad + (13.7 \times 8) + (4.6 \times 15) + (20.1 \times 30) \\
 &\quad + (5.6 \times 18) + (1.5 \times 75) + (14.8 \times 6) \\
 &= 247 + 82.8 + 64 + 220.5 + 273 + 364 + 109.6 + 69 + 603 + 100.8 + 112.5 + 88.8 = 2335
 \end{aligned}$$

$$\sum \%Mass_i = 1.3 + 4.6 + 8.0 + 6.3 + 7.0 + 5.6 + 13.7 + 4.6 + 20.1 + 5.6 + 1.5 + 14.8 = 93.1$$

$$d_{avg} = \frac{2335}{93.1} = 25.1 \text{ km}$$

A weighted average supplier transport distance of 25.1 km was therefore used in the transportation modelling.

A.2 Millipoint (mPt) equivalents

EF3.1	One milli-eco-point equals ...
24.206,7	litres of tapwater from Switzerland
0,9	centimeters road, used for one year
35,9	kilograms of fossil CO ₂ , directly emitted
1,2	kilograms of fossil methane, directly emitted
11,13	grams copper input into agricultural soil
10,8	litres crude oil produced, with transport to the refinery
0,20	grams pesticide application in agriculture
25%	of a person's private daily consumption in Switzerland, 2018
24%	the daily consumption of a person in Switzerland
100,7	km transport of one person by plane
62,5	km transport of one person by car (occupancy 1.6 persons)
1.536,3	km transport of one person by bicycle
102%	of a vegetarian menu with 4 courses
63%	of a meaty 3-course menu
136%	of the daily food consumption of a person in Switzerland, 2018
2,1	plastic carrier bags (production, distribution and disposal)
0,18	cotton T-Shirts
1,2%	of the production of a laptop
335%	of daily consumption for hobbies/leisure activities in Switzerland, 2018
595%	of daily consumption of furniture and household appliances in Switzerland, 2018

Figure 12: Millipoint (mPt) equivalents [8].

A.3 Full comparative table for Scenario 5

Table 15: Scenario analysis for weekly meal impacts

Day	# of Meals	Meals served	Impact	Total impact
Scenario: Status quo week				
Mon.	344	Chicken kebab	0.24	80.97
	113	Mediterranean shakshuka	0.13	14.16
Tue.	119	Oven-roasted celery	0.09	10.68
	293	Cheese tortellini	0.14	41.06
Wed.	269	Beef meatballs	0.53	143.29
	113	Open burrito	0.12	13.37
Thu.	283	Tender roast pork	0.18	50.40
	98	Indian veggie curry	0.14	13.67
Fri.	320	Alaska pollock fillet	0.35	111.13
	86	Cannelloni stuffed with ricotta and spinach	0.10	8.48
				487.20
Scenario: 2 Veggie Days - no pork				
Mon.	344	Chicken kebab	0.24	80.97
	113	Mediterranean shakshuka	0.13	14.16
Tue.	119	Oven-roasted celery	0.09	10.68
	293	Cheese tortellini	0.14	41.06
Wed.	269	Beef meatballs	0.53	143.29
	113	Open burrito	0.12	13.37
Thu.	283	Veggie roast pork	0.10	29.37
	98	Indian veggie curry	0.14	13.67
Fri.	320	Alaska pollock fillet	0.35	111.13
	86	Cannelloni stuffed with ricotta and spinach	0.10	8.48
				466.18
				-0.04
				4% savings
Scenario: 2 Veggie Days - no beef				
Mon.	344	Chicken kebab	0.24	80.97
	113	Mediterranean shakshuka	0.13	14.16
Tue.	119	Oven-roasted celery	0.09	10.68
	293	Cheese tortellini	0.14	41.06
Wed.	269	Plant-based beef meatballs	0.13	36.22
	113	Open burrito	0.12	13.37
Thu.	283	Tender roast pork	0.18	50.40
	98	Indian veggie curry	0.14	13.67
Fri.	320	Alaska pollock fillet	0.35	111.13
	86	Cannelloni stuffed with ricotta and spinach	0.10	8.48
				380.13
				-0.22
				22% savings

Table 16: Scenario analysis for weekly meal impacts (Continued)

Day	# of Meals	Meals served	Impact	Total impact
Scenario: 2 Veggie Days - no fish				
Mon.	344	Chicken kebab	0.24	80.97
	113	Mediterranean shakshuka	0.13	14.16
Tue.	119	Oven-roasted celery	0.09	10.68
	293	Cheese tortellini	0.14	41.06
Wed.	269	Beef meatballs	0.53	143.29
	113	Open burrito	0.12	13.37
Thu.	283	Tender roast pork	0.18	50.40
	98	Indian veggie curry	0.14	13.67
Fri.	320	Veggie alaska pollock	0.08	26.19
	86	Cannelloni stuffed with ricotta and spinach	0.10	8.48
				422.26
				-0.17
				17% savings
Scenario: Vegetarian week				
Mon.	344	Plant-based chicken kebab	0.16	53.43
	113	Mediterranean shakshuka	0.13	14.16
Tue.	119	Oven-roasted celery	0.09	10.68
	293	Cheese tortellini	0.14	41.06
Wed.	269	Plant-based beef meatballs	0.13	36.22
	113	Open burrito	0.12	13.37
Thu.	283	Veggie roast pork	0.10	29.37
	98	Indian veggie curry	0.14	13.67
Fri.	320	Veggie alaska pollock	0.08	26.19
	86	Cannelloni stuffed with ricotta and spinach	0.10	8.48
				246.26
				-0.49
				49% savings

A.4 Complementary Graphs for Packaging

A.4.1 Incineration Scenario

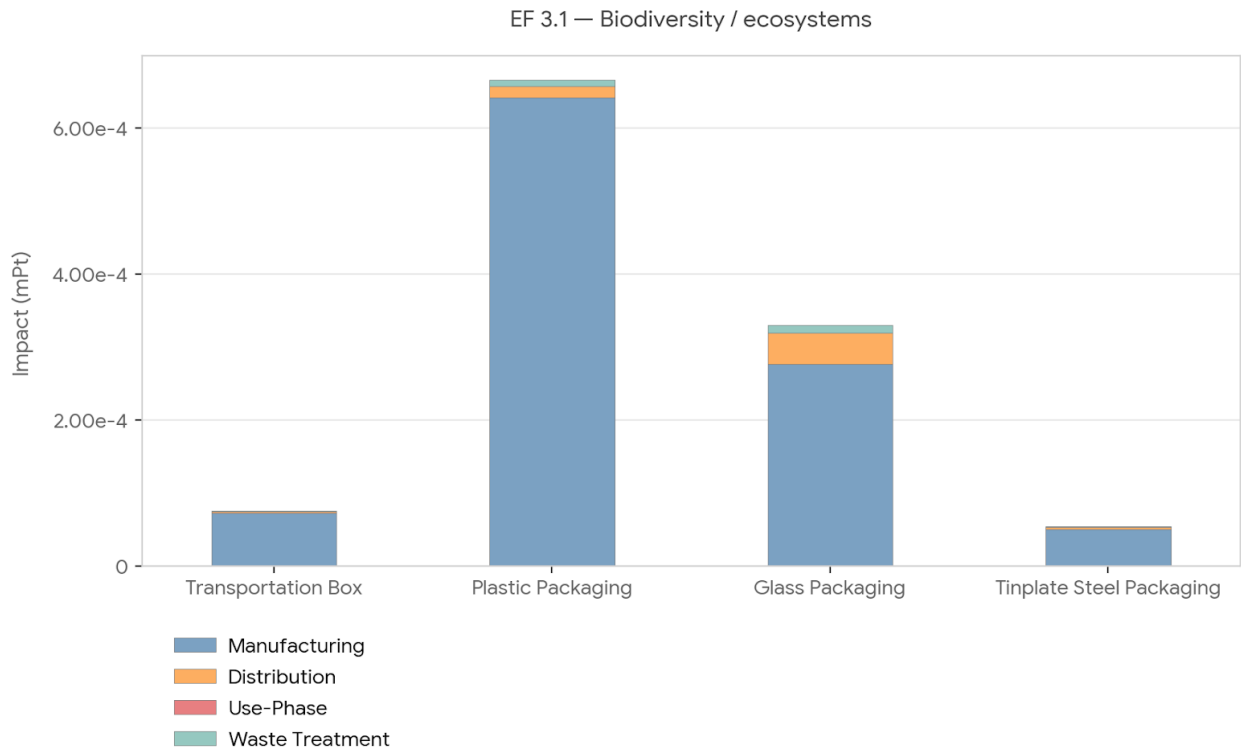


Figure 13: Incineration Scenario: EF 3.1 - Biodiversity / ecosystems

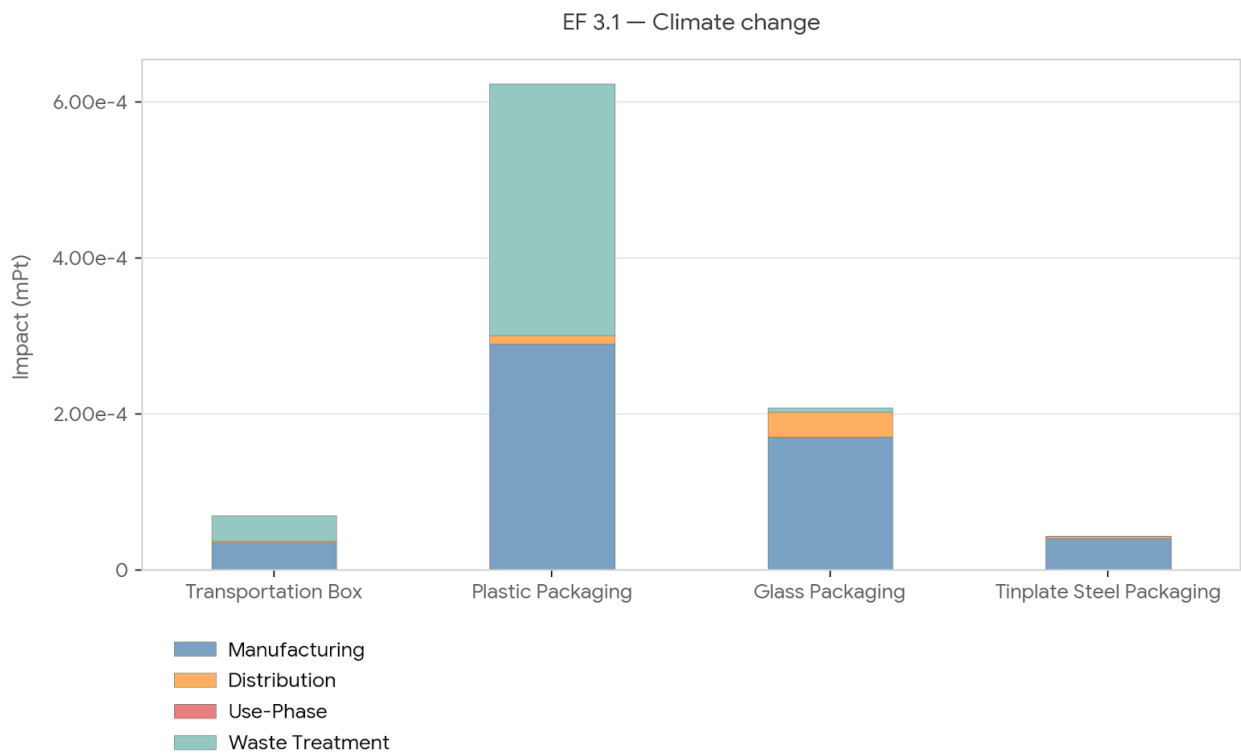


Figure 14: Incineration Scenario: EF 3.1 - Climate change



Figure 15: Incineration Scenario: EF 3.1 - Human Health

A.4.2 Recycling Scenario

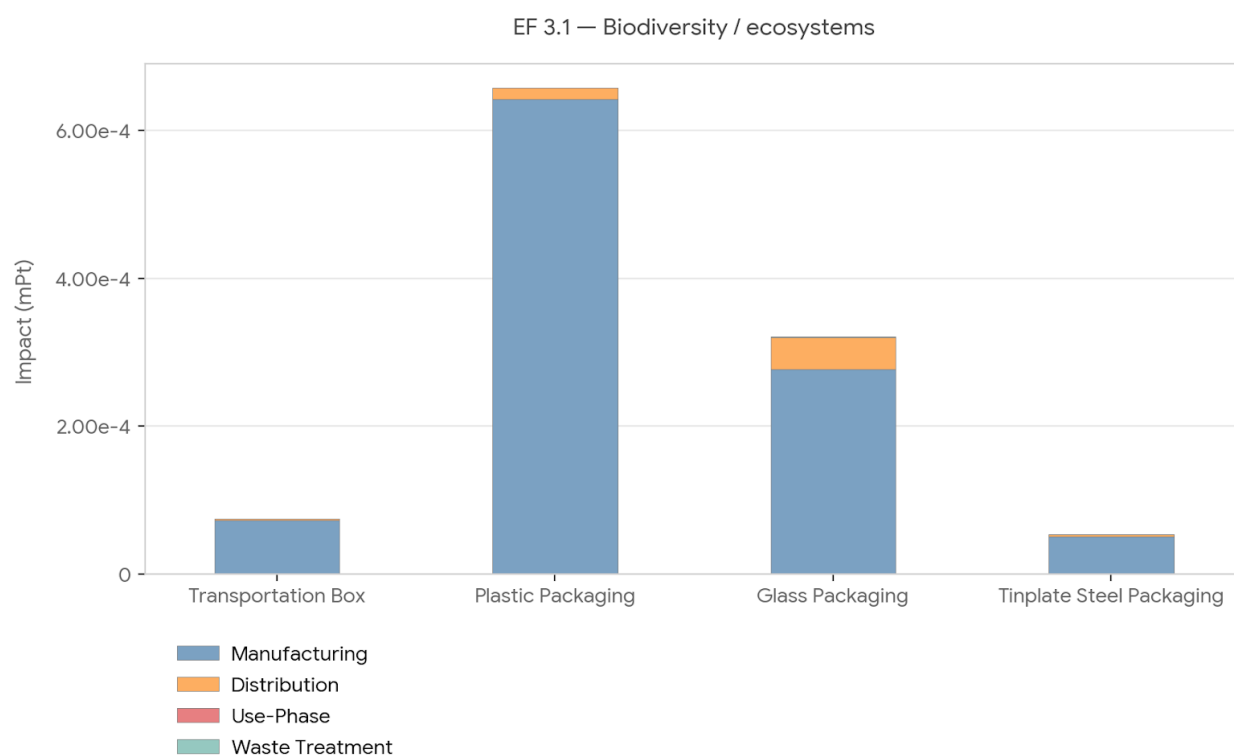


Figure 16: Recycling Scenario: EF 3.1 - Biodiversity / ecosystems

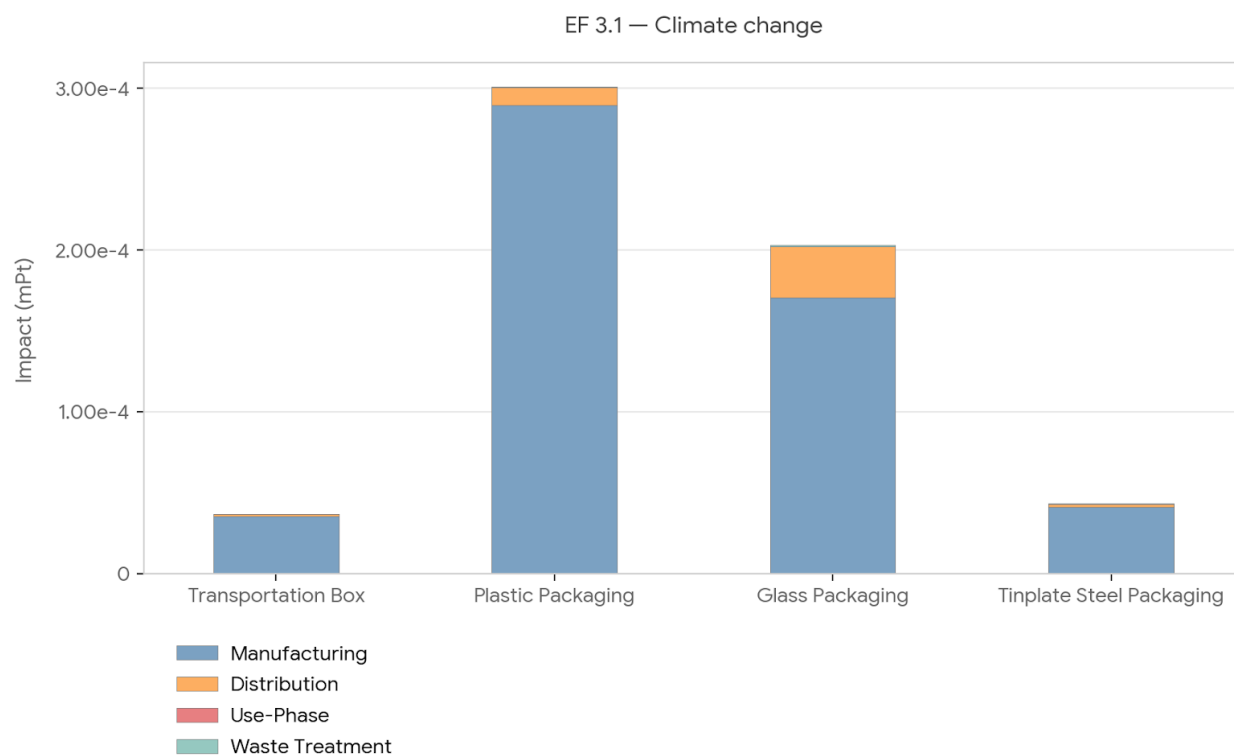


Figure 17: Recycling Scenario: EF 3.1 - Climate change

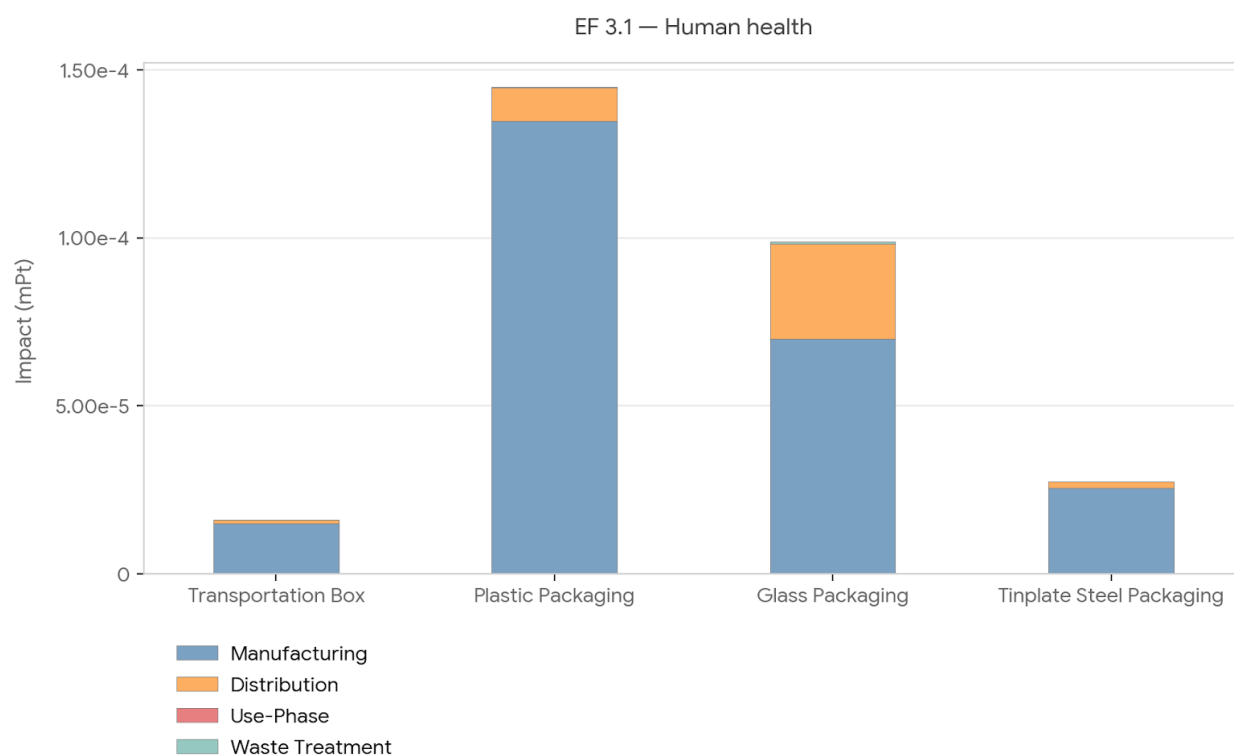


Figure 18: Recycling Scenario: EF 3.1 - Human health

A.4.3 Recycled Materials Scenario

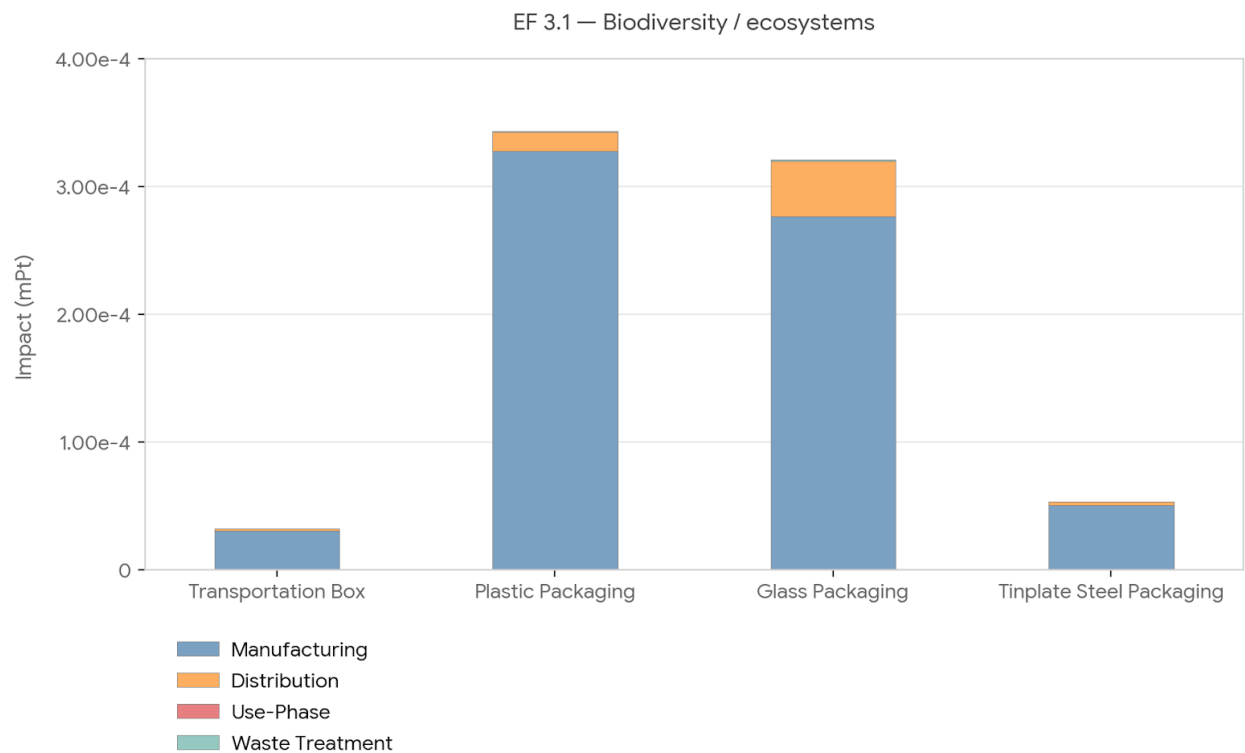


Figure 19: Recycled Materials Scenario: EF 3.1 — Biodiversity / ecosystems

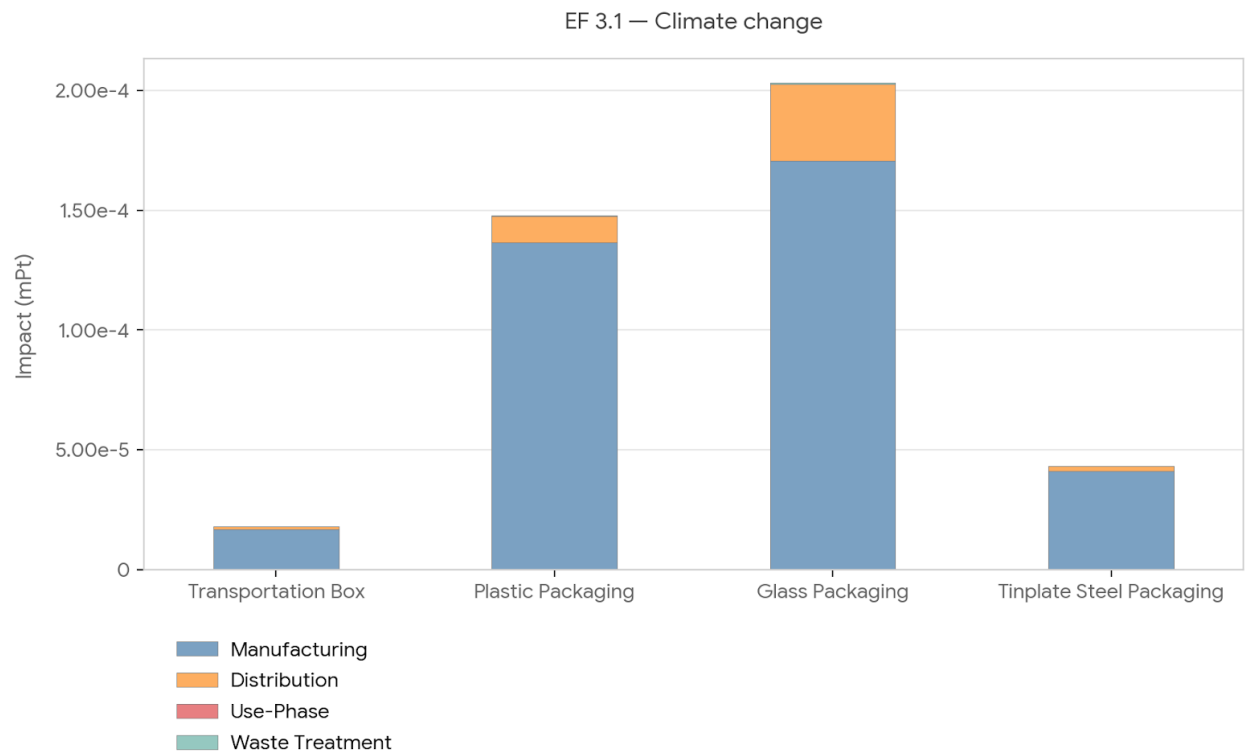


Figure 20: Recycled Materials Scenario: EF 3.1 — Climate change

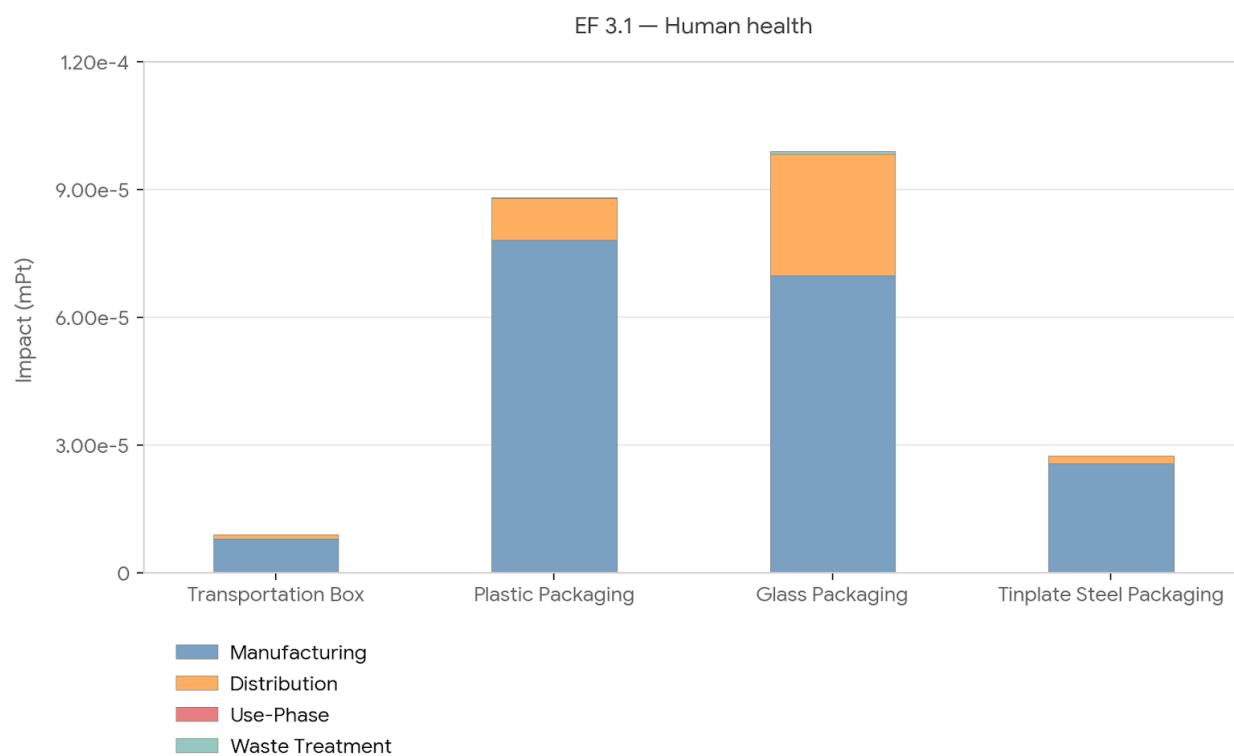


Figure 21: Recycled Materials Scenario: EF 3.1 — Human health